Real-time fusion multi-tier DNN-based collaborative IDPS with complementary features for secure UAV-enabled 6G networks

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A B S T R A C T

UAV-enabled Integrated Sensing and Communication (ISAC) in sixth-generation (6G) wireless networks has sparked significant research interest. UAVs are positioned as aerial wireless platforms, extending coverage and improving Sensing and Communication (S&C) services. However, integrating UAVs introduces vulnerabilities due to multi-connectivity, necessitating robust security measures. To address this, efforts are focused on developing effective Intrusion Detection Systems (IDS). Here ML-based IDSs depend on the most suited Machine Learning (ML) algorithms for improved detection accuracy. However, inadequate detection features frequently contribute to the limitations of detection accuracy in various emerging cyber-attacks. Our study explores attack traits in UAV-enabled 6G networks, using complementary detection features to enhance ML-based attack detection. Additionally, existing Deep Neural Network (DNN) solutions for UAVs-enabled 6G networks mainly use single models (e.g., CNN, SVM, CGAN) instead of multi-tier models, (e.g., Hybrid, Fusion, Ensemble). Here, although the use of a single model can be faster for computer systems, it is difficult to understand the growing complexity of intrusion patterns in data. Also, these single models might not be good at recognizing the unique patterns of less common issues in datasets. Thus, we propose a fusion multi-tier DNN-based Collaborative Intrusion Detection and Prevention System (CIDPS) for critical UAV-enabled 6G networks. This system boosts accuracy without sacrificing latency reductions. Our approach learns decision boundaries from imbalanced data points using preceding DNNs sequentially. Moreover, most existing research works do not focus on intrusion prevention methods and deployment frameworks for UAV networks. This research proposes effective IPS mechanisms and deployment architecture for CIDPS in UAV-enabled 6G networks. It incorporates a CIDPS with an emergency response protocol, neutralizing attacks upon anomaly detection. We validate our solution through diverse dataset experiments (UAVIDS-2020, NF-UQ-NIDS-v2, 5G-NIDD). Unlike traditional practices where IDS are simulated, we implement CIDPS on actual UAV devices (PX4 Vision Dev Kit V1.5, DJI mini se, DJI mini 3 pro) and real-world UAV networks. Our approach outperforms prior algorithms with a 99.25% attack classification accuracy, higher detection efficiency, and lower resource usage, exceeding 99.05% detection rates.

1. Introduction

Following the global deployment and promotion of 5G systems, the focus is shifting towards envisioning 6G networks. The 6G networks integrate sensing, communication, computation and control functions. Similar to their 5G predecessors, 6G wireless networks are anticipated to deliver extensive connectivity to ubiquitously connected devices, each with varying Quality of Service (QoS) demands (Khan, Kumar, et al., 2022). These networks aim to offer ubiquitous coverage, incorporate advanced Artificial Intelligence (AI), optimize energy utilization (Zhao, Huang, Jiang, Gao, & Li, 2022) and bolster adaptive network security (Letaief, Shi, Lu, & Lu, 2021). Since aerial devices are expected to densely populate the airspace, UAVs are poised to play a pivotal role within the 6G ecosystem (Chun, 2023; Waqas, Tang, Abbas, Chen, & Hussain, 2023), serving as an essential network layer that connects networks in space and on Earth. UAVs will be engaged

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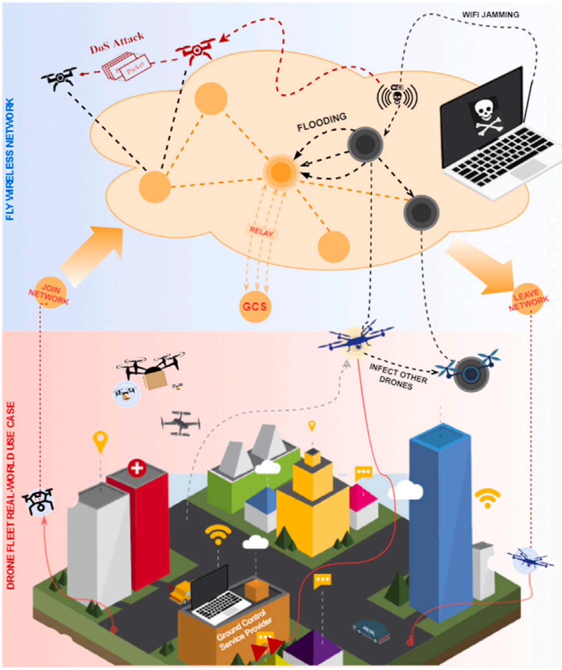

Fig. 1. Components of networked UAVs and potential cyber-attacks.

to communicate with satellites and the ground, establishing a comprehensive space-air-ground network that will enable fully integrated heterogeneous 6G networks (Mishra, Vegni, Loscrí, & Natalizio, 2021).

Likewise, these UAVs can be employed in various capacities, including airborne mobile terminals, aerial base stations, or Access Points (APs), relays, to improve 6G wireless network coverage, dependability, and energy efficiency. Enhanced networks are replacing traditional UAV networks, wherein UAVs establish singular ad hoc connections. Also, UAV networks are frequently implemented in challenging and remote settings, making them susceptible to a range of network cyber emerging threats, i.e., Denial of Service (DoS), flooding and wormhole attacks, as shown in Fig. 1. Therefore, investigating intrusion detection mechanisms is significant for enhancing the security and privacy of UAV networks (Hadi, Cao, Nisa, Jamil, & Ni, 2023). The IDS contributes significantly to increasing the security of UAV-enabled 6G networks by identifying zero-day/unknown attacks (Dang, Amin, Shihada, & Alouini, 2020). However, these attacks are evolving and becoming increasingly inexplicable. Therefore, the research community is now interested in an IDS that relies on Deep Learning (DL) (He, Kim, & Asghar, 2023) and can discover abnormalities. Due to its capacity to automatically identify intricate anomalous behavior and patterns from sizable datasets, DL is suitable for IDS (Kumar, Kumar, Kumar, Sai, & Brahmaiah, 2024). It allows for the detection of new or previously unidentified threats. The main objective of such a system is to increase attack detection accuracy. The following limitations are present in the existing literature.

Overreliance on ML Algorithms: At present, research on IDS uses ML techniques to show their effectiveness. However, the suggested ideas still need a substantial upgrade, because existing literature on boosting detection accuracy concentrates on ML techniques only (Maldonado, Riff, & Neveu, 2022). The improvement of the standard of features employed in a dataset is ignored. These studies used several detection features. Yet, the majority of articles did not explain the selection criteria of the features and their importance, the kinds of attacks they deal with, and the reasons for disregarding the alternative features in the dataset (Fotohi, Abdan, & Ghasemi, 2022; He, Chen, et al., 2023; Saeed et al., 2023; Thakkar & Lohiya, 2023; Wang et al., 2023). For instance, the authors in Wang et al. (2023) highlight that the Transmission Control Protocol (TCP) and Internet Protocol (IP) flags are identified as the most important features. This determination is made after extracting all packet header data and performing dimensionality reduction. Nonetheless, this work did not highlight analysis to uncover other possible traits. In contrast to UAV networks, specific attacks against the UAVs were examined in this article. Thus, to the best of our knowledge, there is still a conspicuous lack of thorough feature selection studies on the UAV-enabled 6G networks, even though current research has examined feature selection in the domains of networks and IoT.

Attack-Specific Characteristics and Data Distributions: Most of the currently available solutions are focused on a single DL model (Abu Al-Haija & Al Badawi, 2022; Alotaibi & Barnawi, 2023; Arthur, 2019; He, Chen, Tang, Wang, & Liu, 2022; Shrestha, Omidkar, Roudi, Abbas, & Kim, 2021). These methods are effective for significantly balanced samples in the dataset and simple attack patterns of various malicious activities. Since new attack techniques have developed, invasive attacks are becoming more varied and sophisticated. Further, the characteristics, tactics and data distribution vary depending on the attack type. As a result, the learning process is intricate and attack classification complexity is rising dramatically. The single DL model also finds it challenging to ascertain some attack patterns due to the availability of limited examples of such attacks (Guo, Li, Liu, Tian, & Kato, 2021; He, Kim, & Asghar, 2023). Under these circumstances, a single DL model method may have trouble understanding attack classes with high prediction performance. Misclassifications are frequent, even when DNNs have undergone intricate hyperparameter tuning.

Inadequate Attention to IPS: Furthermore, the earlier research lacked intrusion prevention techniques and IDPS deployment designs, although they were educational. Several of these studies (Abu Al-Haija & Al Badawi, 2022; He, Chen, et al., 2023; Wang et al., 2023), and He et al. (2022) examined the attack detection process without looking at the response mechanisms following attack detection. Besides, the bulk of these publications suggested a detection model without providing information on how the model is deployed inside the UAV networks architecture (Abu Al-Haija & Al Badawi, 2022; He et al., 2022), and Fotohi et al. (2022). Moreover, there is inadequate prior research on IDPS specifically for UAV-enabled 6G networks.

This paper marks a pioneering contribution in this unexplored domain, presenting the first comprehensive IDPS designed exclusively for the unique challenges and demands of UAV-enabled 6G networks. This study makes several noteworthy contributions, including the following:

- This research introduces a novel methodology, the Proposed Complementary Detection Features (PCDFs), to mitigate the overreliance on machine learning (ML) algorithms in detecting threats to UAV-enabled 6G networks. We create a robust set of features derived from UAV network attacks by meticulously analyzing the nature of potential threats. When these features are integrated with ML techniques, it significantly enhances detection precision. Recognizing the existing literature's limitations, especially in real-time settings, like optimal feature selection, efficient feature extraction, computational constraints, and predictive model characteristics, the PCDFs approach is designed to overcome these hurdles, thereby optimizing detection rates. Our method not only addresses the immediate need for accurate threat detection but also sets a new standard in the adaptability and efficiency of security measures in next-generation network environments.
- To address the challenge of attack-specific characteristics and data distributions, we introduce an approach in the form of a fusion multi-tier DNN model. This model stands out by providing significantly improved intrusion detection accuracy and real-time attack detection capabilities compared to existing solutions. It not only enhances the security posture of high-risk UAV-enabled 6G networks but also plays a crucial role in preventing attacks through its robust attack classification functionality. The development of an advanced IDPS that effectively classifies attacks

and strengthens network security is the main contribution of this work.

- This research introduces a real-time IPS designed to address cyberattacks on UAV networks, effectively remedying the issue of inadequate attention to IPS within existing IDS approaches. By focusing on the prevention and mitigation of cyberattacks across UAV-enabled 6G networks, our proposed architecture employs network virtualization for deploying a CIDPS. This novel approach underscores the practical applicability of the CIDPS in real-world scenarios, highlighting its efficacy in safeguarding UAV networks against a wide array of threats and attacks. The real-time frame destruction capabilities are demonstrated through laboratory-based attack scenarios, utilizing the high-performance Jetson TX2 development board, with the attack traffic generated from an actual PX4 Vision Dev Kit V1.5.

The structure of this research paper is as follows: in Section 2, we explore the related work. We explain possible attacks on UAV-enabled 6G networks, along with the detection of complementary features, in Section 3. In Section 4, fusion multi-tier DNN detection models are illustrated. The design of the proposed real-time collaborative IPS and deployment architecture in UAV-enabled 6G networks is presented in Section 5. Section 6 presents the experimental assessment and discussion. Ultimately, we conclude our research in Section 7.

2. Related work

The research study extensively explores the intersection of three prominent research fields: (1) IDS for UAVs; (2) IDS for UAV Networks; and (3) IDS for 5G/6G Networks. This section provides a comprehensive review of existing works within these aforementioned domains.

2.1. IDS for UAVs

When IDSs are developed for UAVs, Decision Trees (DTs) and Support Vector Machines (SVMs) (Arthur, 2019) are commonly employed for classification tasks. Random Forests (RFs) are extensively employed in ensemble learning. On the other hand, Convolutional Neural Networks (CNNs) (Lu, Xiao, Dai, & Dai, 2020a) specialize in image data analysis, and Recurrent Neural Networks (RNNs) are excellent at analyzing sequence data (Sohn, 2021). These various ML techniques play crucial roles in different aspects of the industry. Recent research emphasized implementing traditional ML methods in IDS design, primarily because of the accomplishments achieved by DL in domains like computer vision (Li & Jung, 2022), speech recognition and Natural Language Processing (NLP). For instance, Elsaeidy, Jagannath, Sanchis, Jamalipour, and Munasinghe (2020) devised a DL-based algorithm for replay attack detection, demonstrating precise discrimination between normal and anomalous behaviors.

Further, the work in Yang et al. (2022) introduced an intrusion detection method. This method combines the Relevance Vector Machine (RVM) with dynamic ensemble incremental learning. The result is an ensemble intrusion detection model that exhibits dynamic adaptability. The work in Arthur (2019) focused on countering signal spoofing and jamming attacks in UAVs. They proposed an IDS based on DL to detect intruders and ensure safe UAV retrieval during central unit disconnection. Their IDS employed Self-Taught Learning and an SVM as a multi-class classifier (Lu, Xiao, Dai, & Dai, 2020b). Another study recommended DL-based IDS for UAVs to detect abnormalities in network traffic (Abu Al-Haija & Al Badawi, 2022). The system performed better than the earlier proposed systems regarding accuracy and effectiveness.

Additionally, the system was accurate in detecting intrusions that were false positives. The recommended solution might not apply to real-world scenarios with diverse and changing network configurations. This is because the system was evaluated using a synthetic dataset. Moreover, writers in Bouhamed, Bouachir, Aloqaily, and Al Ridhawi (2021) suggested a method that uses a compact neural network model. This model was trained and updated regularly to adjust to shifting network circumstances. The scientists employed a network traffic dataset from simulated UAV networks to assess the system performance.

Furthermore, the authors in Basnet, Shash, Johnson, Walgren, and Doleck (2019) used the CIC-IDS2018 and CICIDS2017 datasets to test the generalizability of an IDS dataset. To assess the performance, they deployed twelve learning algorithms from different supervised learning families. A weakness in this study is the assumption that qualitative characteristics in both datasets have an equal amount of new values. A number of abnormalities and threat types, including as GPS spoofing, UAV capturing/spoofing, and routing attacks, are also designed to be recognized by the IDS for UAVs (Al-Haija, McCurry, & Zein-Sabatto, 2021). Recently, there has been a noticeable growth in the creation of cybersecurity solutions based on ML techniques to defend Cyber–Physical Systems (CPS), the IoT, and their connectivity (Rahman & Hossain, 2022).

Moreover, the application of ML classifiers is also presented in Michelena et al. (2023), coupled with PCA, to identify DoS attacks in IoT scenarios, particularly focusing on MQTT protocol networks. The research highlights the effectiveness of classifiers, especially the MLP, which, despite its computational cost, achieved notable accuracy. While this research marks a significant step towards enhancing IoT security, it does not specifically tailor the intrusion detection solutions for the unique challenges of UAV-enabled 6G networks. Also, it does not incorporate the benefits of a multi-tiered system approach, which are critical for addressing the sophisticated threat landscape in such environments.

Additionally, the study in Fu et al. (2023) introduces a novel framework that integrates UAV technology and machine learning to enhance data collection and strengthen the robustness of agricultural information security systems. It utilizes a Double Deep Q-network algorithm for strategic UAV deployment and a comprehensive system for effective intrusion detection. The research also refines intrusion detection with advanced hyperparameter optimization, surpassing traditional methods in accuracy and efficiency.

2.2. IDS for UAV networks

It is necessary to analyze real-time network traffic to discover intrusions targeting UAVs during flight. This can be accomplished by deploying an IDS for UAVs to enable the detection of numerous intrusion classes. Popular intrusion classes include flight signal tampering, routing attacks, malware and message forging attacks (Choudhary et al., 2018). For example, Denoising Diffusion Probabilistic Models (DDPMs) is proposed in Wang et al. (2023) for anomaly detection in UAV network traffic. This IDS achieved high accuracy and low false alarm rates, even in the presence of noise and incomplete data. However, the IDS barely focused on attack classification and handling imbalanced datasets. It is because one-class SVM is trained on a single class of data, which would be normal network traffic. Similarly, a self-adaptive IDS for UAV networks based on the human immune system is proposed in Fotohi et al. (2022). This IDS can detect various malicious attacks in simulated UAV networks, but further evaluation in real-world conditions is needed.

Additionally, in Wang, Wang, Liu, Liu, and Peng (2019), the authors presented the use of a Long Short-Term Memory (LSTM) neural network for UAV anomaly detection. They initially developed a prediction model trained on normal data, allowing them to predict future data points and estimate prediction uncertainty. This method effectively detected point anomalies in real UAV sensor data. Nevertheless, it is worth noting that a singular neural network approach might not be as robust as a hybrid architecture. Combining various methods and techniques could enhance UAV anomaly detection performance. This involves implementing a hybrid framework for improved identification and mitigation of unconventional threats. Moreover, the authors in Da

Table 1
Comparing with the existing machine learning schemes for UAV-enabled 6G networks IDPS.

Ref - Year	Domain: UAV networks(×), UAV-enabled 6G/5G networks (✓) others specified	Dataset	Detection models	Multi-Tier classification	Features selection approaches	Real-time CIDPS (Detection)	Real-in UAV networks environment test
Saeed et al. (2023) - 2023	6G Networks	NSL-KDD, CIC-IDS2017	SVM, RF	×	CS-RF	×	×
He, Chen, et al. (2023) - 2023	×	CIC-IDS2017	Incremental learning	×	None	×	×
Wang et al. (2023) - 2023	×	KDDcup99	DDPM	×	None	×	×
Fotohi et al. (2022) - 2022	×	ToN-IoT	SVMs	×	None	×	×
Fu et al. (2023) - 2023	×	KDD-CUP99	CNN-LSTM	×	None	×	×
Michelena et al. (2023) - 2023	×	Private Dataset	SVM, MLP, GNB, Random Forest	×	None	×	✓
Alotaibi and Barnawi (2023) - 2023	6G Networks	MNIST	IDSoft	×	None	×	×
He et al. (2022) - 2022	×	CIC-IDS2017	CGAN	×	LSTM	×	×
Shrestha, Omidkar, et al. (2021) - 2021	×	CSE-CIC IDS-2018	KNN, LR, DT, K-mean	×	LDA classifier	×	×
Abu Al-Haija and Al Badawi (2022) - 2022	×	UAVIDS-2020	CNN	×	×	None	×
Gawali and Ranjan (2023) - 2023	×	None	Bi-LSTM, DBN	×	Statistical Features	×	×
Ihekoronye, Ajakwe, Kim, and Lee (2022a) - 2022	×	CICIDS2017	Random Forest	×	PCC	×	×
This Work	✓	UAVIDS-2020, NF-UQ-NIDS-v2, 5G-NIDD	CNN, MLP, GAN	✓	Complementary Features	✓	✓

Silva, Ferrão, Dezan, Espes, and Branco (2023) introduced an advanced IDS designed specifically for UAV swarms, with the primary objective of identifying in-flight anomalies and thwarting network attacks. This innovative IDS was developed to tackle the security challenges associated with UAV technology, offering a broader scope of attack detection, ensuring privacy protection, and achieving hardware independence. However, it is important to note a potential gap in the research related to assessing the practical efficiency and real-world impact of the system on hardware components.

In the literature (Ihekoronye et al., 2022a), authors presented an intrusion detection framework for UAV networks, powered by mobile edge computing technology. They used an optimized random forest model on dedicated UAV-MEC (Unmanned Aerial Vehicle-Mobile Edge Computing) servers to efficiently detect various attacks (Zhao et al., 2022). While effective, the reliance on a single DNN algorithm suggests that combining multiple secure methods could further improve attack detection accuracy. Also in literature (Ihekoronye, Ajakwe, Kim, & Lee, 2022b), authors presented an optimized IDS for military operations in Internet of Drones (IoD) networks, using an efficient random forest classifier. It outperformed other models in simulations, but for enhanced accuracy, a mix of secure approaches should be explored instead of relying solely on a single classifier. Besides, the work of Illy, Kaddoum, de Araujo-Filho, Kaur, and Garg (2022a), Illy, Kaddoum, Kaur, and Garg (2022b) introduced the concept of Hybrid DNN-based collaborative IDPS. While their work focused on industrial control systems, it inspired us to explore the applicability of this framework for IDPS in UAV networks enabled by 6G technology. Our proposed architecture leverages a multi-tier structure with complementary DNNs, including autoencoders for anomaly detection and feature selection techniques to enhance the accuracy of intrusion identification. This tailored design, along with the utilization of appropriate datasets for UAV networks, aims to address the unique security challenges faced in this emerging domain.

Additionally, scholars have investigated distributed ML techniques (Saeed, Omri, Abdel-Khalek, Ali, & Alotaibi, 2022), highlighting their potential to reduce the intricacy of high-dimensional data. The benefits of using ML algorithms for pre-processing in IDS to handle vast amounts of data effectively have been highlighted. However, DL algorithms take the analysis a step further by uncovering hidden features during the anomaly classification phase across multiple targets. They consistently exhibit superior performance in terms of false alarms, false negatives, and detection rates, surpassing state-of-the-art technologies while maintaining efficiency, particularly when applied to diverse datasets.

Moreover, recent literature explores various strategies to enhance performance and reduce time consumption, including utilizing parallel processing techniques for accelerated feature selection and extraction (El Aboudi & Benhlima, 2022; Mehanović et al., 2021). The feature selection tasks can be distributed across multiple processing units by leveraging parallel processing, enabling concurrent execution and reducing overall computation time. Moreover, recent advancements in ML and optimization algorithms have led to the development of more efficient feature selection approaches, such as evolutionary algorithms (Bartz-Beielstein, Branke, Mehnen, & Mersmann, 2014) and metaheuristic optimization (Yang, 2011) techniques. These methods aim to improve the speed and effectiveness of feature selection while maintaining or even enhancing prediction performance. These approaches, while valuable, often require substantial computational resources and time, especially when applied to extensive data. In contrast, parallel processing offers a more scalable and time-efficient solution, enhancing the prediction performance while addressing the urgent need for speed in scenarios susceptible to rapid attack vectors, thereby providing a compelling advantage over alternative methods.

2.3. IDS for 5G/6G networks

There is research aimed at secure 6G wireless communication systems to guarantee uninterrupted connectivity for the growing user population and improve ML services. While 6G is in the early stage of development, the standardization of 5G network systems is still ongoing. The optimization of system secrecy rates has been the subject of extensive study, employing various contemporary methodologies. In the literature (Ozpoyraz, Dogukan, Gevez, Altun, & Basar, 2022), the primary objective was to present the latest advancements in DL-based techniques for the physical layer, aiming to pave the way for innovative applications in the context of 6G. Using distributed ML models and wireless communication strategies, the work in Letaief et al. (2021)

presented an idea for scalable and dependable edge AI systems. When the authors introduced a network IDS in Rahman et al. (2022), they suggested an ensemble method using rule learners and decision trees. Their unique DAR ensemble architecture, evaluated on the NSL KDD dataset with unconventional base classifiers, demonstrated experimental accuracies of 80%, 81%, and 15.1% for precision, recall, and false alarm rate, respectively.

Another groundbreaking architecture was introduced by literature (Koursioumpas, Magoula, Barmpounakis, & Stavrakakis, 2022) to proactively detect and forecast anomalies in 5G network traffic. This approach detects anomalies in 5G network traffic automatically by using clustering algorithms and decision tree-based learning. In order to forecast impending network traffic anomalies, they employ a time series model called a BiLSTM autoencoder. Evaluation results indicate the effectiveness and viability of this framework, achieving prediction accuracy rates of up to 90.02%. Further, different ML methods for anomaly detection in 6G networks, including DL, ensemble learning, and one-class SVMs, are explored in Saeed et al. (2023). Ensemble learning techniques enhance accuracy and robustness by combining predictions from multiple ML models. However, they are effective for anomaly detection and are less suitable for classifying specific attacks, particularly when confronted with imbalanced datasets.

Table 1 provides a comparison with the existing ML schemes for UAV-enabled 6G networks IDPS techniques and approaches as documented in the literature. However, the existing literature is based on shallow learning paradigms and is instructive. This means in literature ML models have a limited number of layers, contrasting with deep learning's more complex, multi-layered architectures. The research works cannot fit complex and non-linear distributions for training samples, which are increasing in number. Additionally, these approaches are inappropriate for important UAV-enabled 6G networks. Besides, the particular attention required to high-risk UAV-enabled 6G networks that cannot relinquish security in exchange for faster response times is ignored. Likewise, these studies just propose models for anomaly detection. They ignored the classification of assaults, which is still necessary to stop attacks in the preventive tier. Periodically recommended responses when the attacks are recognized constitute another limitation of the existing literature. Lastly, the literature does not provide insight into how the models in the UAV networks architecture were deployed. Next, the key terminologies used in the literature review including UAV and UAV networks, legacy networks, and 5G/6G systems are summarized below.

UAVs are autonomous aerial vehicles operated without human pilots onboard, whereas UAV networks involve multiple UAVs communicating and collaborating to perform tasks such as surveillance, mapping, or communication relay. Both UAVs and UAV networks offer significant opportunities in various domains including agriculture, disaster management, and infrastructure inspection. Their mobility, flexibility, and ability to access remote or inaccessible areas make them invaluable for data collection, surveillance, and communication. However, there are many challenges in UAV and UAV network deployment including airspace regulations, limited endurance, communication constraints, and coordination among multiple UAVs. Likewise, ensuring reliable communication, navigation, cooperation, and resilience against cyber-attacks in dynamic and uncertain environments remains a challenge.

Furthermore, legacy networks refer to traditional communication networks such as cellular or Wi-Fi networks that were not initially designed to support UAV communication and control. These networks typically lack the capabilities required to meet the stringent requirements of UAV applications. Legacy networks can potentially be leveraged to provide connectivity to UAVs, especially for non-mission-critical applications or low-altitude operations. Nonetheless, integrating UAVs into legacy networks poses challenges such as limited bandwidth, latency, and coverage issues, especially for high-altitude or long-range UAV missions.

Next, 5G and upcoming 6G systems are next-generation cellular communication technologies designed to offer high data rates, low latency, and massive connectivity. These systems promise to revolutionize communication by providing enhanced capabilities tailored to support diverse applications including UAV communication and control. 5G/6G systems hold promise for supporting UAV communication and control with features such as Ultra-reliable Low-latency Communication (URLLC), network slicing, and beamforming. Challenges in integrating UAVs with 5G/6G systems include ensuring compatibility with existing standards, addressing interference and spectrum allocation issues, managing network resources efficiently to accommodate UAV communication requirements, and enhancing security mechanisms to protect against cyber-attacks and vulnerabilities.

3. UAV-enabled 6G networks attacks and proposed detection of complementary features

3.1. UAV-enabled 6G networks attacks

UAVs have a wide range of uses with significant benefits for improving the Quality of Services (QoS). When mobile devices and their users proliferate in UAV networks, intruders decrease the endurance of UAVs by depleting their power through different attacks. The malicious attacks, such as impersonation attacks, jamming attacks and intrusion attacks are described below and shown in Fig. 2.

3.1.1. Jamming attack

Malicious users launch jamming UAVs to interfere with communications in the UAV-enabled 6G networks. The modified UAVs transmit jamming power to prevent UAVs from accessing services or data. Malicious users launch jamming UAVs to interfere with communications in the UAV-enabled 6G networks.

3.1.2. Impersonation attacks

Impersonation attacks can destroy UAV networks security by influencing UAVs (Tao, Han, & Li, 2021). Hackers can control UAVs and act as normal UAVs for delivering services, like sensing and relaying. Malicious UAVs can tamper or steal data during transmission and relay for illegal purposes.

3.1.3. Intrusion attack

Attackers with abnormal intentions might break into UAVs. They can secretly program UAVs with malicious software to collect Personally Identifiable Information (PII). Consequently, users may receive inaccurate information, leading to a low QoS (Tao et al., 2021).

3.1.4. Dos

DoS attacks in UAV-enabled 6G networks can target the communication infrastructure, overwhelming it with a barrage of illegitimate traffic. This flood of data congests channels, rendering them incapable of supporting legitimate commands and data exchanges. As UAVs rely on instantaneous data flow, such attacks can lead to communication breakdowns, hindering mission-critical operations. The potential consequences of DoS attacks on UAV-enabled 6G networks are multifaceted. From a safety perspective, interrupted communications can lead to loss of control, mid-air collisions, or even unauthorized takeovers. Moreover, these attacks could compromise the integrity of data streams, leading to incorrect decision-making and potentially severe consequences.

3.1.5. SkyJack attack

The SkyJack attack takes advantage of the inherent open nature of WiFi communication. An attacker can inject de-authentication packets into the system, disrupting the connection between the authorized operator's smartphone and UAV (Sharma & Mehra, 2023). This disruption effectively cuts off the legitimate operator from controlling the UAV.

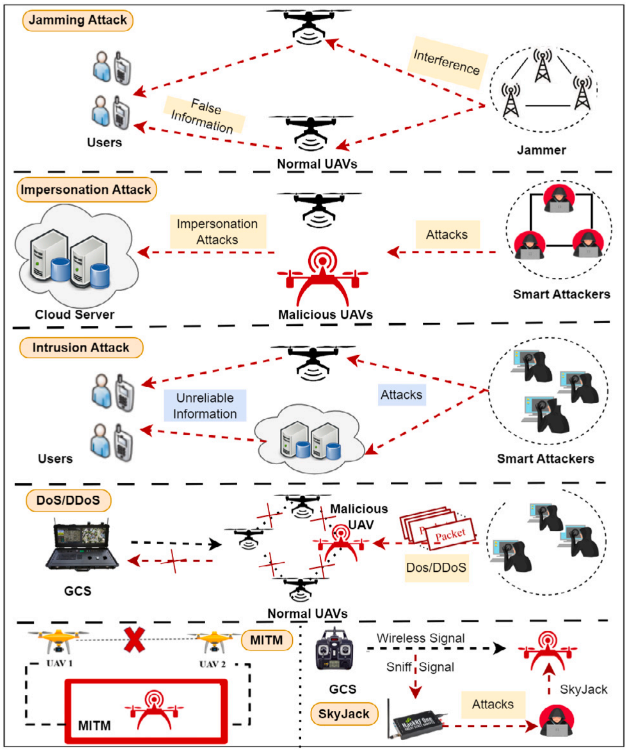

Fig. 2. Intrusion threats/attacks in UAV networks.

3.1.6. Man-in-the-middle (MITM)

In the context of UAV-enabled 6G networks, the threat of MITM attack presents a significant concern. These attacks exploit vulnerabilities in communication protocols, allowing unauthorized intermediaries to intercept data between senders and receivers. As UAVs become integral to 6G networks, the implications of such attacks are amplified. Here, intercepted real-time data transmissions, remote control commands and situational awareness updates can lead to unauthorized, data access and manipulation. It compromises safety, security and the overall reliability of the network.

Further, Table 2 provides a concise yet comprehensive overview of the cybersecurity threats targeting UAV-enabled 6G networks. It details the potential impacts that range from operational disturbance to data breaches, emphasizing the complexity of network security in this emerging field. This synthesized information highlights the critical need for developing sophisticated defense strategies within our research domain.

3.2. Proposed Complementary Detection Features (PCDFs)

One of the most challenging and time-consuming aspects of an ML solution is the selection of a suitable and accurate feature set. The initial feature set is complex for intrusion detection in UAV-enabled 6G networks due to the different components that comprise the system. These factors encompass physical aspects of UAVs and ground stations, configuration, communication protocols, operational principles, workload management and the integration of cutting-edge technologies among others. Therefore, before using dimensionality reduction techniques, a thorough investigation and selecting an initial set of pertinent features is significant. This approach, though challenging, is beneficial as it provides a concise rationale for each feature's selection. The selection of features for ML models in intrusion detection applications primarily depends on the characteristics of the network and the nature of the attacks.

Algorithm 1: Complementary Feature Selection Method For UAV-Enabled 6G Networks IDPS

Input: UAV network attacks $UN\_Attacks$, UAV network features $UN\_Features$
Output: Final detection feature set $Fd\_Features$

Initialize $AS\_Features$ as an empty set ; `// Initialize the feature set`
Initialize an empty list $AS\_Features\_chunks$; `// Initialize list for feature chunks`
Divide $UN\_Attacks$ and $UN\_Features$ into manageable chunks ; `// Divide data for parallel processing`
Start parallel processes for feature selection ; `// Utilize parallel processing for efficiency`
for *each chunk C in* $UN\_Attacks$ **do**
  Initialize $AS\_Features\_chunk$ as an empty set ; `// Initialize feature chunk`
  for *each* $attack\_x$ *in chunk C* **do**
    for *each* $feature\_y$ *in* $UN\_Features$ **do**
      if $feature\_y$ *is encounters by* $attack\_x$ **then**
        Add $feature\_y$ to $AS\_Features\_chunk$; `// Add relevant features`
      end
    end
  end
  Append $AS\_Features\_chunk$ to $AS\_Features\_chunks$; `// Save chunk features`
end
Merge results obtained from each chunk to obtain $AS\_Features$; `// Combine features from all chunks`
`// Enhanced Feature Selection with Dynamic Attack-Feature Representation:`
Initialize $attack\_feature\_matrix$ with dimensions (num_attacks, num_features);
Initialize $attack\_weights$ for each attack type;
for *each* $attack\_x$ *in* $UN\_Attacks$ **do**
  Compute $attack\_weights$ based on attack severity and relevance;
  for *each* $feature\_y$ *in* $UN\_Features$ **do**
    Compute dynamic $feature\_score$ based on susceptibility to $attack\_x$ (consider $attack\_weights$);
    Update $attack\_feature\_matrix[attack\_x][feature\_y]$ with $feature\_score$;
  end
end
Generate synthetic datasets based on $attack\_feature\_matrix$;
Aggregate $attack\_feature\_matrix$ to obtain $AS\_Features$;
Perform dimensionality reduction on $AS\_Features$ if necessary;
Apply feature selection algorithms to select the most informative features;
`// Eliminate Redundant Features:`
Compute the redundancy score for each pair of features in $AS\_Features$;
Sort $AS\_Features$ based on redundancy scores;
Select top features up to a predefined limit;
`// Eliminate Correlated Features:`
Compute the correlation matrix for $AS\_Features$;
Apply a threshold to identify highly correlated feature pairs;
Select one feature from each highly correlated pair;
return $Fd\_Features = AS\_Features$; `// Return the final feature set`

Additionally, the theoretical foundation of our approach is rooted in the foundational principles of UAV network NIDS. Traditionally, NIDSs are designed to monitor and analyze system events for signs of incidents or policy violations. According to Hu, Liu, Chen, Zhang, and Liu (2020), who discuss the game theory of rational detection, a valid intrusion indicator arises from a causal relationship between an attack $attack\_x$ and a measurable change in system state or behavior manifested through system features $feature\_y$ Illy et al. (2022b). This

Table 2
Summary of security threats to UAV-enabled 6G networks.

References	Attack type	Description	Potential impact
Khan, Kumar, et al. (2022)	Spoofing	The forgery of GPS signals to mislead the navigation system of a UAV.	Could lead to unauthorized rerouting and loss of UAV control.
Shrestha, Bajracharya, and Kim (2021)	MitM	Unauthorized interception and possible alteration of communication between two parties.	Compromises confidentiality and integrity of exchanged data, potentially leading to further exploits.
Guo et al. (2021), Tao et al. (2021)	Intrusion	Unauthorized access to network resources, bypassing security mechanisms.	May result in data theft, system damage, or unauthorized control of UAV operations.
Sharma and Mehra (2023)	Skyjacking	Taking control of a UAV by exploiting vulnerabilities in the communication protocol.	Directly jeopardizes UAV operational security and safety.
Guo et al. (2021)	DoS	Overwhelming the network or system with traffic to render it unusable.	Disrupts UAV missions by hindering communication or depleting resources.
Guo et al. (2021)	DDoS	A DoS attack sourced from multiple systems, often involving a botnet.	More severe disruption of services due to the scale and complexity of the attack.
Khan, Sattar, et al. (2022)	Impersonation	An attacker assumes the identity of a legitimate user or device to gain unauthorized access to network resources.	May allow unauthorized access to sensitive areas of the network, leading to data breaches or malicious control.
Shrestha, Bajracharya, and Kim (2021)	Hijacking	Unauthorized seizure of the control signals of the UAV, allowing an attacker to take full control.	May result in a complete takeover, causing the UAV to perform unauthorized operations.

theoretical stance underlies our methodology, wherein we consider a feature affected by an attack if and only if the attack induces a detectable alteration in that feature's behavior or expected values.

In our proposed feature selection method, we operationalize this theory by analyzing features of UAV networks **UN_Features** to discern those with significant changes in their patterns or statistics as a direct consequence of security breaches. This rigorous selection of features ensures that our model is not only empirically but also theoretically grounded in the principles of rational detection. We argue that this approach is well-suited to the context of 6G-enabled UAV networks, where the behavior of devices and network traffic can be distinctly profiled, and deviations can signal potential intrusions.

Furthermore, the theoretical framework of our algorithm, particularly the computation of **attack_weights** and **attack_feature_matrix**, is informed by feature importance theories in ML. This approach is validated by Buczak and Guven (2015) work on contextual ML models, which suggests that the relevance of features in an IDS is dynamic and should be assessed in relation to the type and characteristics of potential attacks.

Considering the behavior of attacks in UAV-enabled 6G networks, this section introduces the PCDFs, and Algorithm 1 illustrates the feature selection process. Initially, the algorithm requires the specification of two principal inputs. The first input is a suite of UAV network attacks **ES_Features**. The second entity is a compendium of network features **UN_Features**. These inputs lay the foundation for the subsequent process that aims to distil an optimized set of detection features **Fd_Features**, ultimately for the identification and prevention of cybersecurity threats. Starting with an initialization phase (line 1), the algorithm formulates an empty set **AS_Features** to collate features of interest. The algorithm tackles voluminous and complex data by employing a strategy of subdivision. This subdivision segments attacks and features into manageable parcels or chunks (line 3). It is a strategic move to utilize parallel processing capabilities, whereby each data chunk is processed together with multiple processing units, significantly accelerating the feature selection phase (line 4).

Next, the core of the algorithm lies in its accurate examination of each feature across every attack chunk, identifying features influenced by specific attack vectors (lines 5–14). This step is crucial for enabling the pinpointing of salient features that are indicative of anomalous or malicious network behavior. The algorithm then unifies these subsets into a comprehensive collection, upon the segregation and identification of pertinent features within each chunk. This collection, identified as **AS_Features** (line 17) embodies the aggregate insight collected from the parallel processing procedures. Advancing to a stage of dynamic assessment, the algorithm formulates an **attack_feature_matrix** and computes corresponding **attack_weights** for each type of attack, reflecting their severity and importance (lines 18–26). This evaluation assigns a dynamic **feature_score** to each feature based on its vulnerability to specific attacks, advocating a contextual and flexible feature selection paradigm. Further, incorporating a dimension of synthetic data generation (line 27), the algorithm utilizes the **attack_feature_matrix** to fabricate datasets, enhancing the pre-existing data corpus. This synthetic data enrichment underpins the robustness of the feature selection framework, enabling its adaptability to diverse and unforeseen attack landscapes.

The algorithm then applies dimensionality reduction where necessary, followed by the application of advanced feature selection algorithms (lines 28–30). These procedures systematically filter the most informative features while concurrently eliminating redundancy and correlation (lines 31–36). Such refinement is crucial to concentrate the feature set to the core constituents, ensuring both computational efficiency and analytical precision for the IDPS. Finally, the process ends with the generation of the final feature set **Fd_Features**, which embodies the crux of the feature selection methodology (line 37). This feature set is a selected collection of features optimally primed for the proactive defense against cyber threats in UAV-enabled 6G networks. The algorithm, with its strategic parallelization and dynamic feature assessment, epitomizes an exhaustive yet streamlined approach to fortifying the security framework of cutting-edge network systems.

3.2.1. UAV-enabled 6G networks traffic features

Features that are evaluated over a set interval of time are referred to as temporal features. An instance of such a feature would be the recognition of distinct connections that, within the past few seconds, typically ranging from three to five, have been made to the same host as the current connection. The predetermined time is five seconds, some attackers employ a significantly wider time than the specified duration set for the IDS. For instance, the occurrence of one connection every 10 s illustrates the efficacy of time-based features. These features are particularly adept at detecting, scanning, and probing activities. The attacks will appear less odd in this situation.

Additional traffic features, specifically those related to connections, are computed within a window defined by a predetermined number of connections. This approach proves effective in overcoming the associated limitations. The quantity of connections within the preceding 60 connections, similar to destination host-like current connections, serves as a connection-based feature. This particular feature contributes to the analysis of network behavior and patterns. In this case, the connection window interval is 60 connections. When attackers deliberately use wider time intervals between their connections to evade detection by IDS systems, these features help in the analysis of UAV networks traffic behavior and the identification of suspicious patterns that indicate an ongoing attack. Selected network traffic features used in the NF-UQ-NIDS-v2 datasets (Sarhan & Layeghy, 2023) are included in Table 3.

3.2.2. UAV-enabled 6G networks packet header features

Packet header features play a crucial role in identifying various network attacks. In most invasive activities, packet headers reveal unexpected values, exposing potential threats like IP fragmentation, SYN flood, MITM, port scanning, and probing in network traffic. For instance, changed TCP flags and inappropriate settings are frequently used in scanning and DoS. Similarly, a MITM attack can also be detected by examining specific responses to Transmission Control Protocol (TCP) flags, notably through the inspection of TCP SYN and TCP SEQ flag responses. In light of this, packet header features can identify these attacks. The packet length can also be used to identify malicious activities. Therefore, an unusual and abnormally large or small data packet is a threat. The 5G-NIDD dataset (Samarakoon et al., 2022) uses packet header attributes listed in Table 4.

The UAVIDS-2020 dataset (Alipour-Fanid et al., 2019) uses packet header attributes listed in Table 5. The PCDFs are summarized in Table 6. No single feature can efficiently detect all threats, they are interdependent and must be combined for training ML models. Different ML algorithms, including basic ML, DL, and ensemble methods, excel in constructing accurate models, but no single algorithm is always the best. Therefore, it is crucial to use advanced intrusion detection for UAV networks, especially in UAV-enabled 6G networks, capable of identifying zero-day attacks like DoS/DDoS, reconnaissance attacks using connection-based and time-based features during training.

Table 3
Selected network traffic features used in the NF-UQ-NIDS-v2 dataset.

Features	Description
FLOW_DURATION_MS	Duration of a network flow in milliseconds.
IN_BYTES	Network traffic feature that quantifies the total number of incoming bytes in a network flow.
OUT_BYTES	Traffic feature that quantifies the total number of outgoing bytes in a network flow.
IN_PKTS	Traffic feature that quantifies the total number of incoming packets in a network flow.
OUT_PKTS	Outgoing number of packets.
MIN_IP_PKT_LEN	Smallest observed length of IP packets within a given network flow.
MAX_IP_PKT_LEN	Largest observed length of IP packets within a given network flow.
MIN_TTL	Captures the smallest observed Time to Live (TTL) value within a given network flow.
LONGEST_FLOW_PKT	Insights into the largest packet size exchanged during a communication session, offering valuable information about data transmission dynamics and potential outliers within the flow.
RETRANSMITTED_IN_PKTS	Count of incoming packets that have been retransmitted within a network flow.
L7_PROTO	Identifies the specific protocol that applications are using to exchange data.

Table 4
Selected of network packets header features used in 5G-NIDD dataset.

Features	Description
IPV4_SRC_ADDR	The source IPv4 address in a network communication session.
IPV4_DST_ADDR	The destination IPv4 address in a network communication session. This address specifies the intended recipient of data packets.
TCP_FLAGS	Cumulative of all TCP flags.
CLIENT_TCP_FLAGS	TCP flags observed on the client-side of a network flow.
SERVER_TCP_FLAGS	TCP flags observed on the server-side of a network flow.
ICMP_TYPE	ICMP message in a network flow.
DNS_QUERY_ID	The identifier assigned to a DNS query in a network flow. DNS query IDs help ensure the correct association between queries and responses.
DNS_QUERY_TYPE	DNS query type (e.g. 1=A, 2=NS..).

4. Fusion multi-tier DNN detection models (FMDDM)

In this section, we present the components and development phases of the proposed fusion multi-tier DNN model. The learning dataset for this model is created by utilizing both training and validation datasets. The test dataset is subsequently employed to evaluate the performance of the final fusion multi-tier DNN. In the development of this model, a set of DNN algorithms, namely multi-layer Perceptron

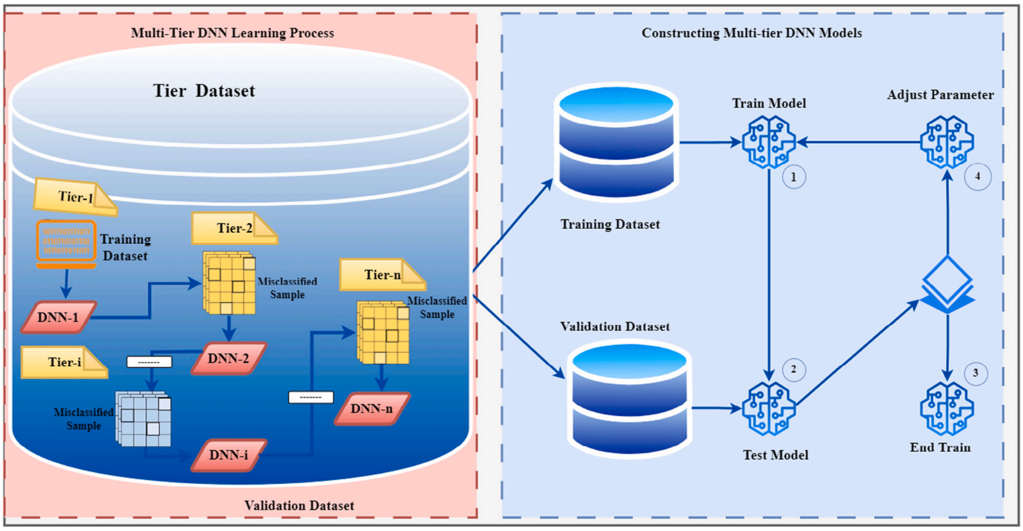

Fig. 3. Activity diagram: Constructing multi-tier DNN Models.

Table 5
Selected of network packets header features used in UAVIDS-2020 dataset.

Features	Description
Uplink_size_MAX	It provides information about the maximum amount of data uploaded from a device to a network during a communication session.
Uplink_interval_MIN	The data transmission from a drone or UAV to a central control station or other communication endpoints.
Uplink_size_Skewness	Quantifies the skewness of the distribution of data sizes in the uplink direction of a communication session. Skewness is a measure of the asymmetry of a probability distribution.
Uplink_size_Kurtosis	kurtosis of data size distribution in the uplink direction of a communication flow.
Downlink_interval_Skewness	Data transmission from a central control station or server to a drone or UAV.
Both_links_size_STD	The variability or dispersion of data sizes exchanged between devices, encompassing both data sent from the UAV to the control station (uplink) and data sent from the control station to the UAV (downlink).
Downlink_interva l_MeanSquare	The square root of the average of the squared time intervals between successive downlink transmissions.
Downlink_size_MAX	The data transmitted from a central control station or other communication endpoints to a UAV.

(MLP), Convolutional Neural Network (CNN), and Generative Adversarial Networks (GAN), serve as the primary elements. The development of the model has three main phases. The first tier is a learning phase, which encompasses training, validation, and hyperparameter tuning. The second tier is the evaluation phase, where the model's performance is rigorously assessed. Lastly, the third tier is the prediction phase, during which the model is applied to make predictions on new or unseen data.

Table 6
Proposed complementary detection features.

Attack category	Attack type	Traffic features	Header features
Reconnaissance	Scanning Fuzzers	Host and port discovery Abnormal patterns	Probe requests Modified payload
Exploitation	Exploits Shellcode Backdoor	Unusual traffic spikes Unusual code execution Covert data transmission	Exploited vulnerabilities Malicious payload execution Encrypted commands
Credential Theft	Password Theft Brute force	Unusual login patterns Multiple login attempts	Stolen credentials Repeated authentication requests
Malware	Worm Bot Ransomware	Rapid propagation Unusual behavior Encryption and ransom note	Replication mechanism Command and control traffic Ransom message inclusion
Web Attacks	XSS MITM	Injected malicious scripts Interception and alteration	Altered input parameters Manipulated data flow
Denial-of-Service	DoS DDoS	Traffic overload Massive traffic influx	Abnormal request frequency Coordinated source IPs
Generic	Generic	Varied and unpredictable	Various attack-specific markers
Evasion	Analysis	Peculiar analysis patterns	Sandbox detection evasion

4.1. Datasets

4.1.1. UAVIDS-2020

The IDS model is trained to detect cyber-attacks in UAV networks using the UAVIDS-2020 dataset. It includes non-redundant, current Wi-Fi traffic logs that have been encoded. Additionally, all logs list binary output tags as 0 for typical UAVs and 1 for anomalous UAVs. The collection also includes other metadata in addition to testing and training records. Logs for the computational creation of each feature and the computational dependency logs between different features are included in the metadata (Alipour-Fanid et al., 2019). The UAVIDS-2020 dataset is available to the general public as MAT files containing various unique, non-repeated samples. This distinction increases the detection procedure's accuracy since frequent logs better protect the binary classifier. The dataset can also identify complicated UAV traffic patterns and online threats. It has applicable characteristics and computational statistics for classifying UAV traffic into two categories. The dataset, featuring authentic network traffic data from three widely recognized UAVs (Parrot Bebop, DBPower UDI, and DJI Spark), inherently validates its real-world applicability.

4.1.2. NF-UQ-NIDS-v2

The NF-UQ-NIDS-v2 dataset presents an extensive collection of network traffic, encompassing both normal and malicious activities, as detailed in Sarhan, Layeghy, and Portmann (2022) research. Updated to include the most recent attack patterns, this dataset is apt for the development of machine learning models aimed at intrusion detection within 6G networks. It comprises 75,987,976 entries, with 25,165,295 (33.12%) representing normal traffic flow and 50,822,681 (66.88%) associated with cyber attacks.

4.1.3. 5G-NIDD

The 5G-NIDD dataset serves as a resource for network intrusion detection, constructed from data gathered on an actual 5G testbed. Released in 2022 by Samarakoon et al. (2022) it encompasses 52 attributes which include 32 floating-point, 12 integer, and 8 categorical types. The dataset contains a total of 1,215,890 entries, with benign instances numbering 477,737, making up 39.2% of the data, and malicious instances totaling 738,153, constituting 60.7%. It represents the types of traffic that will be extended to next-generation wireless networks, such as 6G networks. This data can be used to train and evaluate intrusion detection systems for UAV-enabled 6G networks. It can also be used to develop new intrusion detection techniques specifically tailored to the unique characteristics of these networks.

4.2. Training phase

The multi-tier DNN stages are developed in order. Using various DNN models and the learning dataset referred to as the tier_1 dataset, we train various DNN models. The training phase of each DNN is illustrated in Fig. 3.

After the first tier of learning is over, the trained models are used to categorize samples in the tier_1 dataset. The model with the highest accuracy is DNN-1. The second tier dataset, i.e., the tier_2 dataset, contains the misclassified samples or exhibited low confidence in the classification results of DNN-1. During the second tier, multiple DNN algorithms and the tier_2 dataset are employed to retrain diverse DNN models. The training process shown in Fig. 3 is the same.

The tier_2 dataset samples are classified using the trained models once the second learning tier is completed and the top model is saved as DNN-2. In a dataset for the following step, incorrectly categorized data or misclassified samples with low confidence are collected for the successive tier dataset. The subsequent tiers are developed step-by-step until the quantity of incorrectly identified samples decreases below a predefined threshold. Algorithm 2 presents the training algorithm for this fusion of multi-tier DNN.

Initially, Algorithm 2 ingests an initial dataset comprising both normal and attack samples, alongside a set of predefined DNN algorithms, various thresholds, and parameters necessary for the training process. The training commences at the first tier and progresses iteratively through subsequent tiers. In each tier, cross-validation is employed to split the dataset into training and validation sets, which are then normalized to ensure consistent feature scaling. The normalization process is crucial for preventing the model from developing a bias towards features of a particular scale. The core of the algorithm is SAE and DNN models within each tier. The early stopping mechanism plays a pivotal role here, it monitors the validation loss during the SAE training and the validation accuracy during the DNN training. If there is no improvement observed over a defined number of epochs, the training is halted. This strategy ensures that the model does not memorize the training data, thus enhancing its generalization capabilities. After training, the algorithm evaluates each sample in the current tier's dataset using the trained DNN model. Samples that are either misclassified or classified with confidence below a predefined threshold are identified. These samples make the dataset for the next tier, allowing the model to focus specifically on samples that were challenging to classify in the previous tier.

Algorithm 2: Training the Fusion Multi-Tier DNN Framework

Input: Initial Dataset containing Normal and Attack samples.
DNN_Algorithms: Predefined DNN algorithms.
Min_confidence: Predefined confidence threshold.
Min_dataset: Predefined threshold for new tier creation.
Θ: Predefined threshold for DNN performance improvement.
Patience: Number of epochs to wait for improvement before early stopping.

```
Initialize Tier_X to 1
while Data Size of Tier_X_dataset > Min_dataset do
    foreach Fold of Cross_validation do
        Split the dataset into Training (80%) and Validation (20%) sets.
        Normalize the dataset: (x−min(x)) / (max(x)−min(x))
        Divide the samples into four sets, ensuring a balanced distribution with
          50% Normal and 50% Attack samples in each set.
    end
    SAE Model Training:
    Initialize patience_counter to 0
    Initialize best_loss to ∞
    foreach epoch do
        foreach balanced dataset I do
            Train the autoencoder: minimize L(x_i, x_i) using BCE loss and
              Adam optimizer.
            Calculate validation loss on the validation set.
            if validation_loss < best_loss then
                best_loss ← validation_loss
                patience_counter ← 0
                Save the model weights
            end
            else
                patience_counter ← patience_counter + 1
            end
            if patience_counter ≥ Patience then
                Break Early stopping
            end
        end
    end
    DNN Model Training:
    foreach balanced dataset I do
        foreach Predefined algorithms for DNNs do
            Initialize patience_counter to 0
            Initialize best_accuracy to 0
            while patience_counter < Patience do
                Train and validate DNN, iterating on hyperparameters for
                  enhanced performance < Θ.
                if current_accuracy > best_accuracy then
                    best_accuracy ← current_accuracy
                    patience_counter ← 0
                    Save the model weights
                end
                else
                    patience_counter ← patience_counter + 1
                end
            end
            Load the best model weights
            Choose DNN-X as the DNNs with the highest validation accuracy.
            Initialize an empty table Misclassified_samples.
        end
    end
    foreach Sample in Tier_X_dataset do
        Employ DNN-X to make predictions for the sample class.
        if Misclassified by DNNX or Certainty < Min_confidence then
            Misclassified_Samples ← Sample
        else
            Tier-X ← Tier-X + 1
            Misclassified_Samples ← Tier-X dataset
        end
    end
end
```

Result: Trained fusion multi-tier DNN $\{DNN1, DNN2, \ldots, DNNn\}$ and the final fusion multi-tier model.

Also, the algorithm iteratively refines the model through each tier, focusing on challenging samples and employing early stopping to ensure that the model does not overfit to the training data. This tiered approach, coupled with early stopping, systematically enhances the model's robustness and accuracy, particularly for hard-to-classify instances. The final output is a series of trained DNNs across multiple tiers, culminating in a comprehensive fusion multi-tier model that is adept at classifying both normal and attack samples in the dataset.

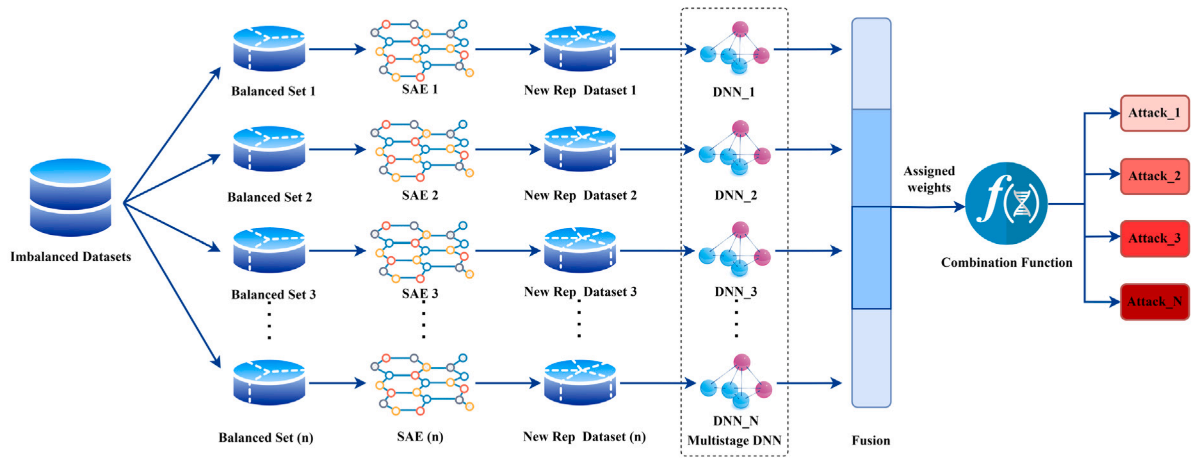

Fig. 4. Evaluation workflow using fusion multi-tier DNN.

Table 7
Combination rules for fusion multi-tier DNNs integration.

Combination Rules	Function Mapping $f(x)$
Maximum	$Y = \max_{j=1}^{N} d_{ji}$
Minimum	$Y = \min_{j=1}^{N} d_{ji}$
Median	$Y = \text{median}(d_{1i}, d_{2i}, \ldots, d_{Ni})$
Sum	$Y = \sum_{j=1}^{N} d_{ji}$
Weighted Sum	$Y = \sum_{j=1}^{N} w_j d_{ji}$

4.3. Assessment and prediction phase

All test dataset samples are processed and assigned weights during the assessment phase. The suggested approach involves creating a well-balanced representation from an initial raw dataset, which is then employed as input for a fusion multi-tier DNN model for classification. Within the fusion multi-tier DNN architecture, multiple unsupervised Stacked Autoencoders (SAE) are utilized to extract new feature representations from datasets that have an uneven distribution of classes. The SAE-based attack detection model leverages multiple Autoencoders (AE) to derive fresh representations from unlabeled data, capturing diverse patterns. Subsequently, the feature representations derived from each Stacked Autoencoder are conveyed to a DNN using super vectors and amalgamated within a multi-tiered fusion DNN framework. As seen in Fig. 4, the decision on fusion multi-tier DNNs is subsequently created using these assigned weights in conjunction with other functions. The utilization of the accuracy metric in evaluating our model's performance is predicated on the deliberate structuring of the data feeding into our Fusion Multi-Tier DNN. As represented in Fig. 4, the model's design incorporates the division of input data into multiple balanced sets, each comprising an equivalent number of normal and attack instances. This symmetry in the datasets across all tiers mitigates the common evaluation challenges associated with accuracy in skewed datasets. The accuracy metric provides a genuine reflection of the model's discriminative power, by ensuring that the volume of attack samples is proportionate to the volume of normal samples at each stage of evaluation. Table 7 outlines the different combination functions used to classify the test dataset. These functions include simple sum, maximum, minimum, median and weighted sum.

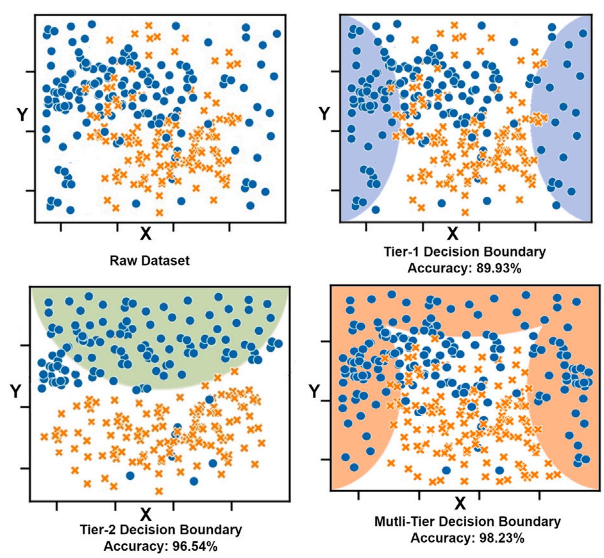

Fig. 5. Multi-tier DNN Benefits: Analyzing decision frontiers across stages.

The Y in this case is the fusion multi-tier DNN output for the instance i. It is computed as the fusion multi-tier DNN weighted sum of the outputs from each DNN, with the weights based on how well each DNN performs on the training set. The output of $DNNs_j$ for the instance i is denoted by d_{ji} in this case, and N shows total DNNs used in fusion multi-tier DNN. The weights $w_j d_{ji}$ are determined by training the multi-tier DNN on a labeled dataset of network traffic data. On the test dataset, the multi-tier DNN is given the combination function, which results in the best classification accuracy. Lastly, the proposed approach is implemented in UAV-enabled 6G networks to categorize incoming network traffic.

Fig. 4 determines the same procedure for new incoming traffic. Using the method (fusion multi-tier DNN), misclassified samples at each level are reused to boost the DNN's capacity for learning at the following tier rather than being wasted. Therefore, each successive tier model focuses on a distinct data subset. It discovers a novel optimal decision frontier that classifies incorrectly classified samples from previous tiers. The models also learn about the distribution of minority classes during the subsequent tiers of DNN. In short, the fusion multi-tier DNN architecture presented in Figs. 3 and 4 operates under a

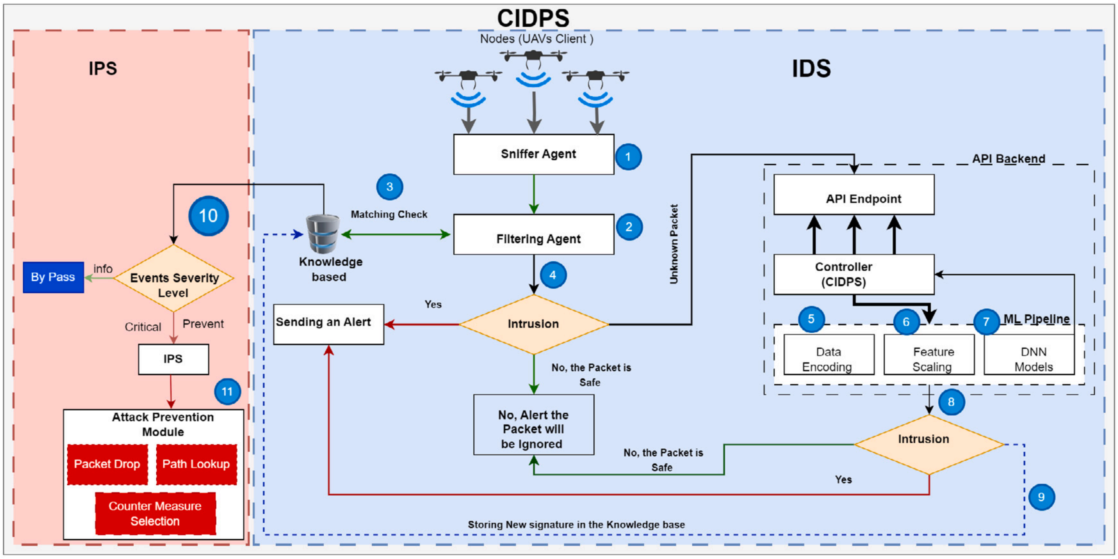

Fig. 6. Real time proposed collaborative IDPS architecture in UAV-enabled 6G networks.

unified training and validation framework. Fig. 3 illustrates the model's iterative training loop, encompassing model training, validation, and hyperparameter tuning. Complementarily, Fig. 4 expounds on the pre-processing and feature extraction pipeline through SAEs, leading into the tiered training structure where misclassified samples from one tier are input into the subsequent tier. The cohesive training strategy, employing a consistent loss function across tiers, ensures that the model is effectively trained on both balanced and imbalanced datasets, catering to the diverse and evolving nature of threats in UAV-enabled 6G networks.

Meanwhile, Fig. 5 illustrates the improved accuracy mechanism of the proposed approach with three tiers of decision frontiers. The first tier represents the original dataset, achieving an accuracy of 89.93%. The second tier showcases the decision frontier after the first tier DNN, reaching an accuracy of 96.54%. The third tier displays the decision frontier after the multi-tier DNN, attaining an accuracy of 98.23%.

5. Proposed real-time collaborative IDPS and deployment architecture in UAV-enabled 6G networks

This section outlines the preventive measures taken upon detecting an attack. Additionally, it details the proposed deployment architecture for CIDPS in UAV networks.

5.1. Proposed real-time collaborative IDS

The real-time CIDS in the UAV networks is essential for immediate response and efficient threat mitigation. In CIDS, the term "Sniffer Agent" refers to each unit component of the system. A monitoring unit or both can analyze each CIDS "Filter Agent". The network traffic is captured and logged by a monitoring component. A high-level overview of how to propose a CIDS for UAV networks in real-time is given in Fig. 6. The functional block diagram illustrates the overall flow of the whole system. The gateway router and the UAV networks are set up in the Linux environment. The packet sniffing technique examines the network traffic passing via the Linux Ground Control Station (GCS) system. Consequently, the features are extracted from the acquired data.

The extracted data is organized as a feature array by the controller of the Linux environment. The data is sent as a Hypertext Transfer Protocol (HTTP) request to API endpoint of the API backend over the internet. The ML pipeline receives data extracted from the HTTP request via the API backend controller. There are two different modules of an ML pipeline, i.e., categorical data encoding and feature scaling. The first component of the ML pipeline converts categorical data into numerical values. Contrarily, the second component scales the entire dataset to a standard scale. The pre-processed data is input into the trained DNNs model. In response to the associated HTTP request, the API backend controller returns the prediction outcome as an HTTP response. Also, trained fusion Multi-Tier DNN optimized for real-time anomaly detection and attack classification, are deployed to the API backend. This system receives network traffic data from the packet sniffing component, and upon detecting suspicious patterns, generates alerts forwarded to the controller. Operating as the central command hub, the controller processes these alerts, potentially logging incidents, notifying administrators, and executing corrective measures. Moreover, it seamlessly interfaces with the IPS, enabling actions such as dropping, rejecting, or allowing traffic based on the attack classification of the detected anomalies to proactively fortify network security.

5.2. Proposed real-time collaborative IPS

Likewise, an advanced countermeasure is employed when critical attacks of utmost severity are identified. In such cases, the controller directs the system to intercept packets destined for specific IP addresses associated with the attack. These intercepted packets are then deliberately dropped, preventing any potentially harmful data from reaching its intended target. This multi-layered approach to network security ensures that common attack flows are swiftly neutralized, but critical threats are also proactively addressed, bolstering the network's overall resilience against malicious activity.

Within the proposed IPS for UAV-enabled 6G networks, encapsulated in a holistic framework outlined in Algorithm 3, the operational structure involves initiating an assessment based on the policy upon the arrival of a packet, determining whether to accept or drop it (lines 1–3). The packet is discarded promptly if the matched policy yields a "drop" verdict. Conversely, if the policy permits, the packet is admitted, and the system proceeds to create or update internal session states and session stateful attributes (lines 3–4).

Subsequently, the system constructs comprehensive features for the packet incorporating derived session features. The system employs a Classification and Prioritization Controller (CPC) to ascertain the benign or malicious nature of the packet's associated session. If the classification score surpasses the predefined Malicious Classification Score (MCS) threshold, the session is either included in the policy's blacklist or whitelist, depending on its classification (lines 8–17). If the

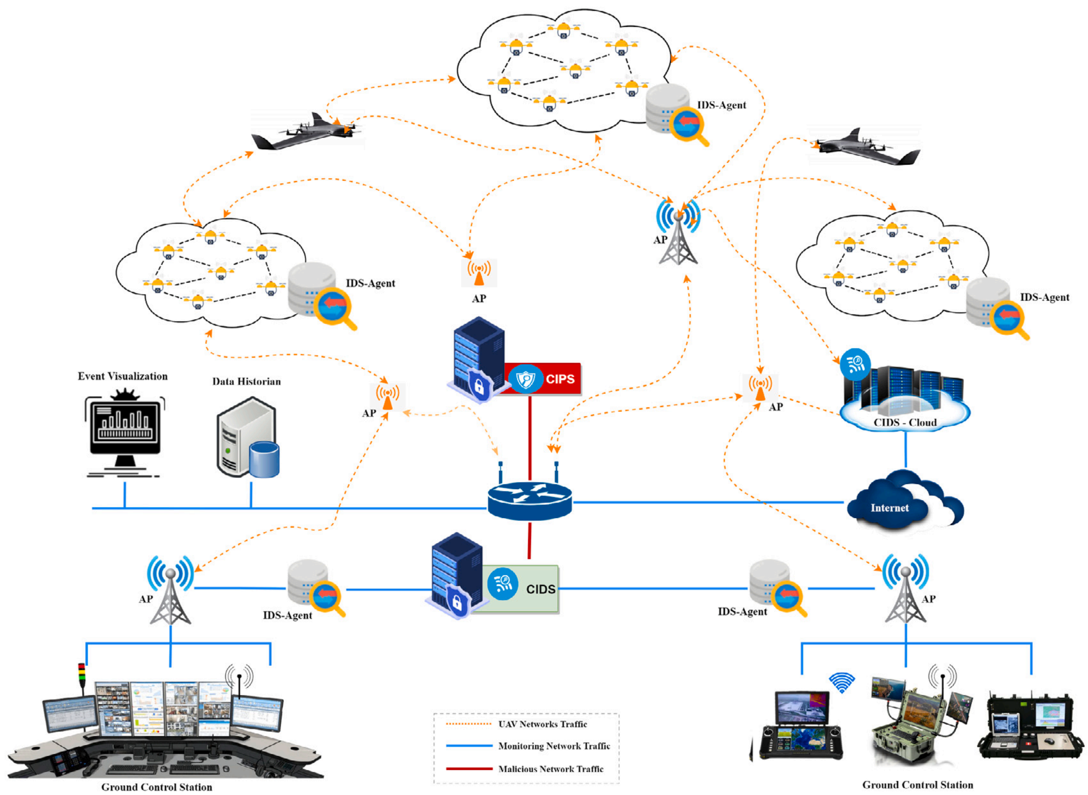

Fig. 7. Real time proposed collaborative CIDPS and deployment architecture in UAV-enabled 6G networks.

Algorithm 3: Intrusion Prevention System (IPS)

```
   Input: Critical Alerts & Events
   Result: Drop Malicious Packet
1  if Packet P received then
2      if contents in IPS for repository P then
3          if content POLICY is "drop" then
4              Drop P ;                  // reset bi-directional session
5              Notify;
6              Return;
7          else
8              if content POLICY is "alert" then
9                  Forward P;
10                 Notify;
11                 Return;
12             end
13         end
14     else
15         Classify packet P using CPC;
16         if CPC score < MCS (Malicious Classification Threshold) then
17             Return;
18         end
19     end
20     if contents NOT in IPS for repository P then
21         Match P with IDPS_ML Model with timeout T;
22         if P is Malicious then
23             Ask User to define POLICY;
24             if POLICY is "drop" or POLICY is "alert" then
25                 Update IPS Repository;
26                 Update POLICY Table;
27             end
28         end
29     end
30 end
```

score is lower than the MCS threshold, the packet is forwarded without considering its classification. Upon the conclusion of the session, the internal session state and session stateful feature data expire and are purged once the final features have been compiled. The conclusive decision regarding the session is made by the Traffic Flow Controller (TFC), with the results logged and, if deemed necessary, relayed to the administrator. If the packet contents are not cached, the algorithm matches the packet with the IDPS_ML model. The IDPS_ML model is an ML model that is trained to detect malicious packets. The algorithm also sets a timeout T. If the model takes longer than T seconds to respond, the algorithm drops the packet (lines 20–21).

Next, if the packet is malicious, the algorithm asks the user to define a policy for the packet. The user can choose to drop the packet, alert the administrator, or take no action (line 23). If the user chooses to drop or alert the administrator, the algorithm updates the IPS repository and policy table. The IPS repository stores information about known malicious patterns, and the policy table stores the policy for each known malicious pattern (lines 24–26).

5.3. Deployment architecture of CIDPS in UAV-enabled 6G networks

The suggested deployment architecture for CIDPS in residential networks is shown in this section, along with the preventative steps are taken when an attack is identified as shown in Fig. 7. This study introduces a collaborative fusion multi-tier DNN architecture to address the time-critical security challenges in high-risk UAV-enabled 6G networks, further enhanced by including a Cloud-based IDS agent. The initial component, known as UAV-CIDS, operates locally on a server and specializes in anomaly detection through binary classification, allowing for the immediate triggering of emergency responses upon anomaly detection. Subsequently, the second multi-tier DNN component, UAV-cloud CIDS, assumes responsibility for classifying these anomalies by their attack type, thereby identifying appropriate complementary response measures. By harnessing the swiftness of binary classification for initial detection, this architecture empowers high-risk UAV-6G networks to respond swiftly and mitigate the impact of attacks. If additional computational resources are required, processing anomalous traffic indicates just a fraction of the entire network data, the cyber attacks classification module (UAV-Cloud CIDS) can be easily implemented on cloud servers to ease the workload.

This strategy augments the responsiveness of the IPS without compromising the classification accuracy of the IDS. The proposed IDPS deployment model adopts a tailored approach for UAV-6G networks, ensuring effective implementation. In this model, multiple UAVs are interconnected within the UAV networks, forming a robust communication ecosystem. We receive data from UAV networks through agents, with these agents strategically deployed within the range of all UAV networks. The agent is tasked with capturing and sending network traffic data to a CIDS via a local server equipped with the fusion multi-tier DNN-based anomaly detection module (UAV-cloud CIDS).

Furthermore, remote fields with similar setups may be interconnected. When UAV-CIDS notices an abnormality, it notifies the IPS promptly and sends pertinent traffic data. The IPS creates emergency response policies, which are then transmitted to the relevant components of the UAV-capable 6G network for automatic configuration changes, therefore strengthening network resilience. Additionally, UAV-Cloud CIDS classifies the attack features it receives from CIDS before forwarding the relevant traffic information and the classification. Subsequently, the IPS generates supplementary response rules, which are then implemented on the network infrastructure. Advanced modules can also be deployed for countermeasures, including tactics to slow down the attacker's system, enhancing the overall security posture of UAV-enabled 6G networks.

6. Experiment

This section is dedicated to our experimental assessment and the subsequent discussion of the obtained results. The main goal is to evaluate the suggested fusion multi-tier DNN models for effectiveness. The attack classification and anomaly detection modules employ these models.

6G network datasets enabled by UAVs typically have certain drawbacks, notably the lack of plausible attack scenarios consisting of multiple event nodes that might occur sequentially. We conduct our experiments on three different datasets, each with a unique attack combination, classification complexity and data distribution to validate our approach. These datasets are carefully chosen to ensure a robust evaluation of our proposed methodology.

In our experimental evaluation, initially, we implemented three different approaches and then compare them, (i) Single Deep Neural Networks (DNNs), (ii) Multi-tier DNNs with varying hyperparameters at each tier, and (iii) Fusion multi-tier DNN. These approaches utilize DNN algorithms, namely MLP, CNN, and GAN. The two main phases of our experimental process are attack classification and anomaly detection. With the help of the well-known open-source ML library scikit-learn, the experiments are conducted in the Python programming language.

6.1. Anomaly detection

In the initial tier, we implement and assess the anomaly detection module using three different datasets: UAVIDS-2020, NF-UQ-NIDS-v2, and 5G-NIDD. The organization of these datasets for binary classification is outlined in Table 8.

For the UAVIDS-2020 dataset, we adopt an 80%–20% split strategy, dividing it into tier_1 for learning (80%) and testing (20%). Within each tier, the learning dataset underwent further partitioning. This process involved creating training (60%) and validation (40%) subsets, allowing for effective fine-tuning of our models. Regarding the NF-UQ-NIDS-v2 dataset, we employ it as our primary learning dataset. For the 5G NIDD dataset, we similarly use an 80%–20% split, with 80% data designated for learning and 20% for testing.

Fig. 8 illustrates the model accuracy achieved on the UAVIDS-2020 dataset, while Fig. 9 showcases the model performance on the NF-UQ-NIDS-v2 dataset. Fig. 10 illustrates the model accuracy achieved on the 5G-NIDD dataset. Concerning the fusion multi-tier DNN model, the reported accuracy is ascertained by identifying the optimal combination of functions used during the evaluation process.

Table 8
Arrangement of UAVIDS-2020, NF-UQ-NIDS-v2 and 5G-NIDD datasets for binary categorization.

Class	Dataset Record Count		
	UAVIDS-2020	NF-UQ-NIDS-v2	5G-NIDD
Normal	42, 136	25,165,295	477,737
Attacks	26, 600	50,822,681	738,153
Total	68, 736	75,987,976	1,215,890

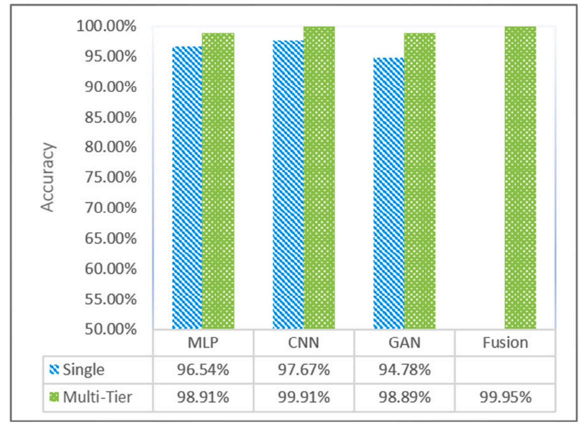

Fig. 8. Evaluating the performance of FMDDM anomaly detection models on the UAVIDS-2020 dataset.

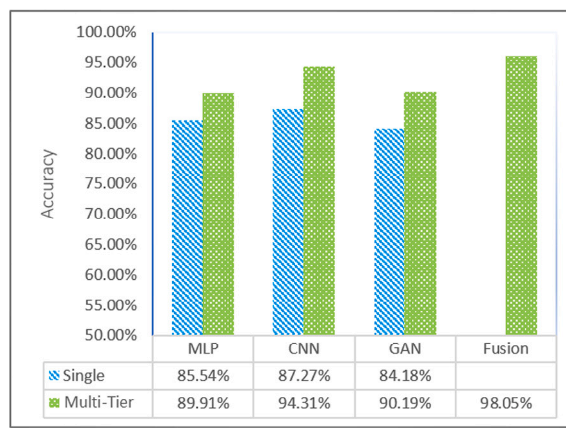

Fig. 9. Evaluating the performance of FMDDM anomaly detection models on the NF-UQ-NIDS-v2 dataset.

The fusion multi-tier DNN also outperforms other DNNs using the UAVIDS-2020 datasets, with a detection accuracy of 99.95%, as shown in Fig. 8, an undetected rate of 0.23%, and a false alarm rate of 0.01%. The simple multi-tier DNN performs noticeably worse in this dataset, achieving less than 98.91% detection accuracy. Additionally, the single DNN performance was notably poor, achieving an accuracy of less than 94.78%.

Moreover, using the NF-UQ-NIDS-v2 dataset, the fusion multi-tier DNN indicates exceptional performance, achieving a detection accuracy of 96.05%, as presented in Fig. 9. Additionally, it demonstrates an exceptionally low false alarm rate of less than 0.19% and a remarkable undetected rate of only 1.18%. It also shows an outstanding missed detection rate (false negative rate) of only 1.18%.

Furthermore, it showcases an impressively low false alarm rate (false positive rate) of less than 0.19%. It also exhibits a remarkable undetected rate (false negative rate) of only 1.18% and an impressively

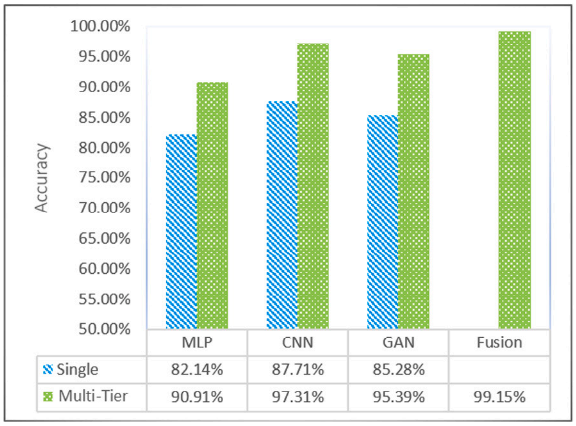

Fig. 10. Evaluating the performance of FMDDM anomaly detection models on the 5G-NIDD dataset.

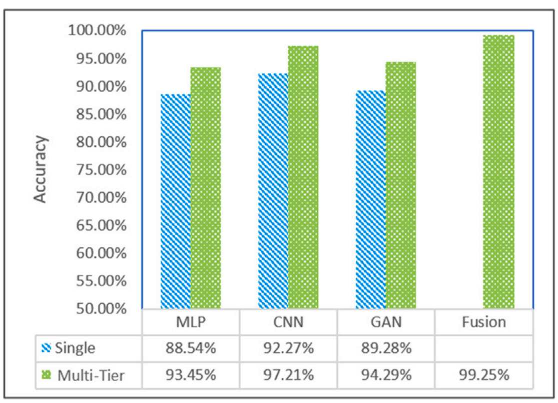

Fig. 11. Attack classification model accuracy on NF-UQ-NIDS-v2 + 5G-NIDD dataset.

Table 9
Weights for each DNN model on different datasets, determined by detection accuracy.

Model type	Detection accuracy	Calculated weight
Table 10a: UAV-IDS 2020		
Fusion Multi-Tier DNN	99.95%	0.3404
Simple Multi-Tier DNN	98.91%	0.3368
Single DNN	94.78%	0.3228
Table 10b: NF-UQ-NIDS-v2		
Fusion Multi-Tier DNN	98.05%	0.3528
Simple Multi-Tier DNN	94.31%	0.3394
Single DNN	85.54%	0.3079
Table 10c: 5G-NIDD		
Fusion Multi-Tier DNN	99.15%	0.3559
Simple Multi-Tier DNN	97.31%	0.3493
Single DNN	82.14%	0.2948

low false alarm rate (false positive rate) of less than 0.19%. Furthermore, all the basic multi-tier DNNs achieve an outstanding detection accuracy of 94.31%, while the individual single DNN exhibit strong performance with detection accuracies exceeding 85.54%. Maintaining a single DNN to comprehend the intricate data distribution of intrusion patterns in the NF-UQ-NIDS-v2 dataset is challenging. The dataset poses a significant difficulty for models. This dataset consists of 49 distinct attacks. Moreover, using the 5G-NIDD, the fusion multi-tier DNN demonstrates exceptional performance, achieving a detection accuracy of 99.15%, as indicated in Fig. 10.

Additionally, it shows an exceptionally low false alarm rate of less than 0.19% and a remarkable undetected rate of only 1.18%. A system of multi-tier DNNs achieved an outstanding detection accuracy of 97.31%, while the single DNN exhibits strong performance with detection accuracies exceeding 82.14%. It asserts that comprehending the intricate data distribution of intrusion detection patterns in the 5G-NIDD dataset, which encompasses 19 distinct attacks, proves challenging for models relying on a single DNN. It highlights the complexity inherent in this particular task.

Besides, the fusion multi-tier DNN is more efficient at learning intrusion patterns than the models that used standard multi-tier DNN. The dataset's infiltration patterns involve less intricate data distribution than for practical UAVIDS-2020 in actual UAV networks. Therefore, both single-tier DNN and simple fusions multi-tier DNNs demonstrate commendable detection accuracy. It illustrates that creating accurate intrusion detection models for UAV networks, severely at risk of different attacks, is a challenge. These models cannot perform well by using just a single DNN. Besides, it demonstrates that fusion multi-tier DNN is particularly effective in creating more reliable intrusion detection models than single-tier DNN. Table 10 provides a comparison of anomaly detection model performances across three datasets; UAVIDS-2020, NF-UQ-NIDS-v2, and 5G-NIDD. It details the detection accuracy, FNR, and FPR for Fusion Multi-Tier, Simple Multi-Tier, and Single DNN models. Notably, Fusion Multi-Tier DNNs consistently show high accuracy and low FNR and FPR across all datasets, significantly outperforming Single DNN models which exhibit higher FNR and FPR. The Simple Multi-Tier DNNs perform between these two extremes, offering a trade-off between the high efficiency of Fusion models and the lower performance of Single models.

Next, Table 9 a, b and c provided below exhibit calculated weights corresponding to each DNN model based on their detection accuracy. For instance, the Fusion Multi-Tier DNN shows a higher detection accuracy, and consequently, the calculated weight is comparatively greater, indicating a stronger reliance on this model's outputs. Conversely, the Single DNN, with the lowest accuracy, is assigned the least weight, signifying less influence in the final decision-making process.

6.2. *Attacks classification phase*

Attack classification is the focus of the second part of the experiments. We only perform attack classification on the NF-UQ-NIDS-v2+ 5G-NIDD datasets since the UAVIDS-2020 dataset does not include the attack type. The NF-UQ-NIDS-v2 + 5G-NIDD dataset for classifying different attacks is shown in Table 8. This NF-UQ-NIDS-v2+5G-NIDD is used as the test dataset after we divide the learning dataset, NF-UQ-NIDS-v2+5G-NIDD Train+, into training and validation datasets. The models for attack categorization are constructed using identical DNN architectures, encompassing single-tier DNN, simple multi-tier DNN, and fusion multi-tier DNN. This uniformity ensures consistency across the various model configurations. On the NF-UQ-NIDS-v2 + 5G-NIDD dataset, Fig. 11 validates the accuracy of the model. The reported accuracy for the fusion multi-tier DNN model comes from using the best combination rule functions.

The results of the attack classification demonstrate that a single DNN is less accurate than a multi-tier DNN. Furthermore, compared to the single-tier DNN, the proposed fusion multi-tier DNNs detect the attacks with higher accuracy. The proposed IDPS, based on fusion multi-tier DNNs, delivers predictions with superior accuracy. This outperforms previous literature that relies on different datasets for anomaly detection and attack classification. Table 11 presents a comparison of the attack classification model (which has twenty-two categories) and the accuracy used in previous studies. The effectiveness

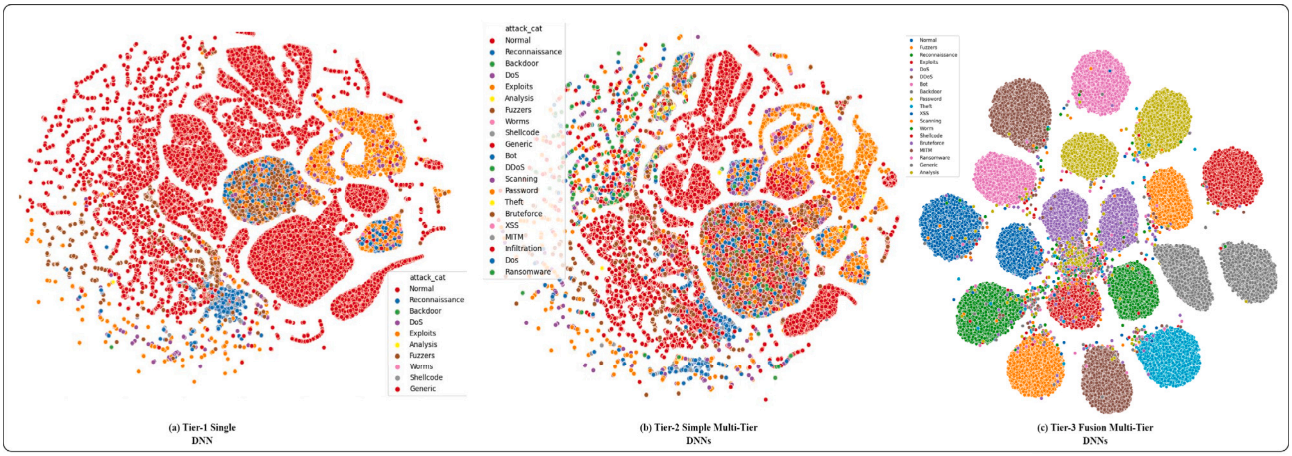

Fig. 12. Attacks visualization result of NF-UQ-NIDS-v2, and 5G-NIDD.

Table 10
Performance summary of FMDDM detection models on various datasets.

Dataset	Model type	Detection accuracy	FNR	FPR
	Fusion Multi-Tier DNN	99.95%	0.23%	0.01%
UAVIDS-2020	Simple Multi-Tier DNN	98.91%	1.13%	0.17%
	Single DNN	94.78%	4.20%	6.45%
	Fusion Multi-Tier DNN	98.05%	1.18%	0.19%
NF-UQ-NIDS-v2	Simple Multi-Tier DNN	94.31%	5.46%	9.23%
	Single DNN	85.54%	12.65%	19.21%
	Fusion Multi-Tier DNN	99.15%	1.08%	0.16%
5G-NIDD	Simple Multi-Tier DNN	97.31%	2.06%	4.12%
	Single DNN	82.14%	16.85%	22.91%

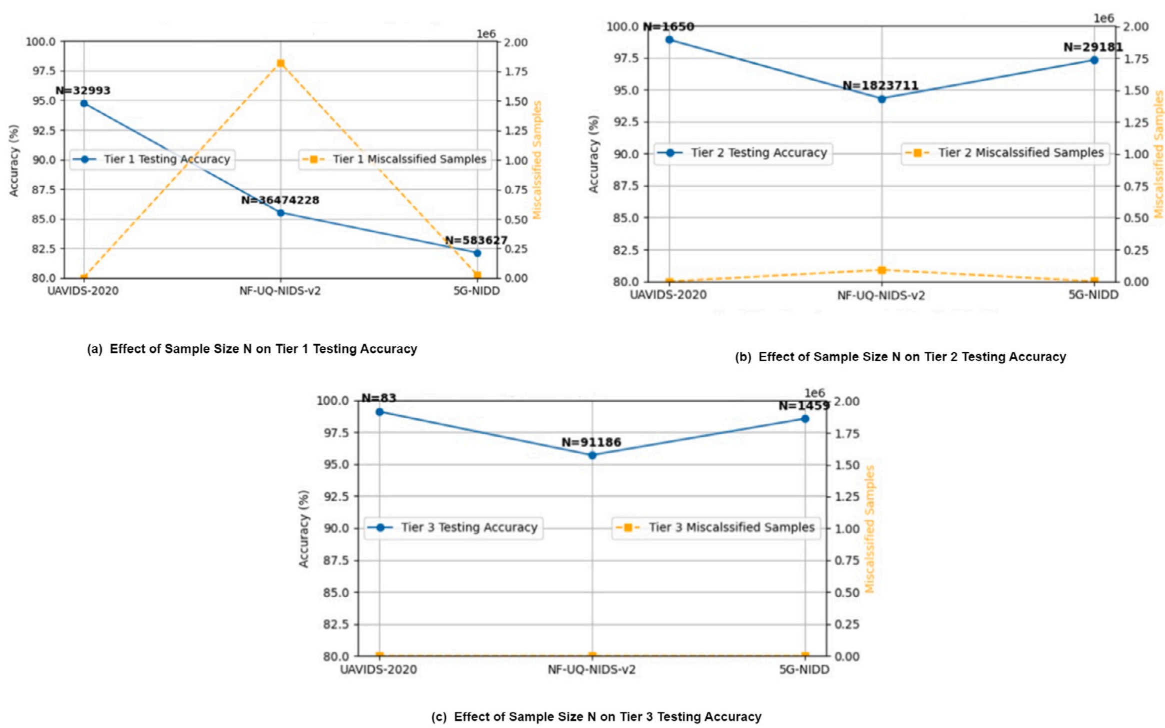

Fig. 13. Effect of sample Size N on Tier 1, 2, 3 testing accuracy across datasets.

of the suggested approach is demonstrated by the experimental evaluation, i.e., more effective for important UAV-enabled 6G Networks with security as their main objective.

Our study examines the original NF-UQ-NIDS-v2 and 5G-NIDD datasets, where no attacks are present. Employing t-distributed Stochastic Neighbor Embedding (t-SNE) visualization, we visually depict the initial data distribution to establish a baseline for normal network traffic. Subsequently, we introduce a simple DNN-based intrusion detection model to the dataset, allowing us to visualize the alteration in data distribution (Tier 2), as shown in Fig. 12. Here, we witness the emergence of distinct clusters representing various types of detected attacks. These clusters exist alongside overlapping points that reflect normal traffic.

Table 11
Structure of the NF-UQ-NIDS-v2, and 5G-NIDD dataset for cyber attacks classification.

Class	Dataset record count	
	NF-UQ-NIDS-v2	5G-NIDD
Reconnaissance	263,3778	
Backdoor	18,978	
DoS	178,75585	
Exploits	31,551	
Analysis	2299	
Fuzzers	22,310	
Worms	164	
Shellcode	1427	
Generic	16,560	
Bot	14,3097	
DDoS	217,48341	
Scanning	37,81419	
Password	11,53323	
Theft	2431	
BruteForce	12,3982	
XSS	24,55020	
MITM	7723	
Infiltration	11,6361	
Ransomware	3425	
UDPFlood/ HTTPFlood/ SYNFlood/ICMPFlood		609,028
Slowrate DoS		73,124
TCPConnect Scan/ SYN Scan/UDP Scan		56,001

While this marks a significant step forward in intrusion detection, our research reaches its zenith in the third tier—a fusion multi-tier DNN-based IDS. As illustrated in t-SNE Tier-3, this multi-tier approach refines the data distribution even further, showcasing a notable separation of attack clusters from normal traffic. This visual validation underscores the efficacy of our fusion multi-tier DNN-based intrusion detection system in achieving improved classification accuracy for various attack classes within the NF-UQ-NIDS-v2 and 5G-NIDD datasets.

Additionally, the fusion multi-tier DNN model's performance, when trained on varying sizes of datasets, reveals a critical understanding of the sample size's impact on intrusion detection accuracy. Tier 1's results, as visualized in Fig. 13(a), indicate that the model's detection capabilities initially improve with the number of samples. This trend is particularly evident when comparing the results from the UAVIDS-2020 dataset with a relatively modest sample size to the expansive NF-UQ-NIDS-v2 dataset, which showcases a substantial influence on the testing accuracy and number of misclassified samples. When the model progresses to Tiers 2 and 3, the impact of increased sample size becomes more distinct, highlighting a sophisticated balance between data volume and detection refinement. Figs. 13(b) and 13(c) illustrate that despite the immense volume of samples provided to Tier 1, the testing accuracy in subsequent tiers does not always show proportional improvement. This observation is crystallized in Table 12, which compiles the model's performance metrics across all tiers, offering an aggregated view that correlates the number of samples to testing accuracy and misclassification rates. These insights are instrumental in optimizing our fusion multi-tier DNN model, ensuring it uses the full potential of the provided datasets while maintaining high detection accuracy.

6.3. Intrusion prevention performance over time

The classification of network intrusions by our CIDPS is intrinsically tied to the subsequent prevention mechanisms. In our system, once a potential threat is identified and classified by the IDS component,

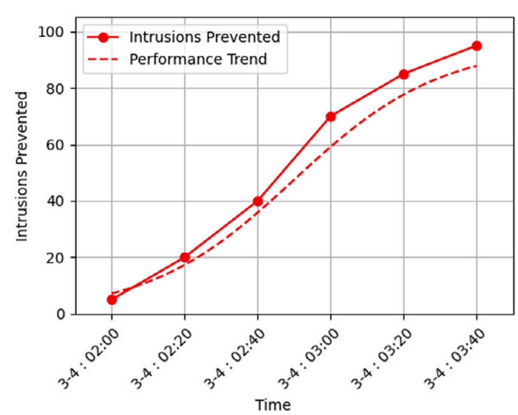

Fig. 14. Increasing intrusions prevented over time.

this triggers the IPS component to take preventive actions, which can include blocking the attack, isolating the affected network segment, or other countermeasures. In Fig. 14, the x-axis represents time, while the y-axis quantifies the intrusions thwarted by the UAV network. Initially, the graph indicates that only a few intrusions are detected and averted, suggesting the system's nascent stage of detecting potential threats. As time advances, the system's ability to prevent intrusions is enhanced, denoted by the ascending curve on the graph. The increasing trend illustrates the system's evolving efficacy, culminating in a more fortified environment under its protection. This progression evidences the system's capability to adapt and improve, fulfilling its role as an effective IPS over time.

6.4. Real-word CIDPS implementation

Deploying the proposed CIDPS in UAV networks, with the NVIDIA Jetson AGX Orin Developer Kit as the host system, presents unique

Table 12
Performance summary of a multi-tier classification model across different datasets.

Dataset	Total samples	Tier 1 samples	Tier 1 testing accuracy	Misclassified samples tier 1	Tier 2 samples	Tier 2 testing accuracy	Tier 3 samples	Tier 3 testing accuracy
UAVIDS-2020	68,736	32,993	94.78%	1650	1650	98.91%	83	98.91%
NF-UQ-NIDS-v2	75,987,976	36,474,228	85.54%	1,823,711	1,823,711	94.31%	91,186	94.31%
5G-NIDD	1,215,890	583,627	82.14%	29,181	29,181	97.31%	1459	97.31%

Table 13
Specifications of UAV networks devices and Jetson TX2's.

Device	Description	Specification
UAV1 (PX4 Vision Dev Kit)	Processor	NXP i.MX 8M Quad core ARM Cortex-A53 CPU
	GPS	U-blox ZED-F9P GPS module with concurrent GNSS
	Optical Flow Sensor	PMW3901 connected via Telem3 Serial
	Operating System	PX4 Autopilot
UAV2 (DJI Mini SE)	Processor	MediaTek MT6761E
	Operating System	DJI Fly App
	Dimensions	138 mm × 80 mm × 58 mm
UAV3 (DJI Mini 3 Pro)	Processor	MediaTek MT6779
	Operating System	DJI Fly App
	Dimensions	1459067 mm
GCS 1	Operating System	Windows 11
GCS 2	Operating System	Android
GCS3 (RadioMaster TX16S)	Channel	16 channels
	Radio firmware	OpenTX
Jetson TX2	CPU	Nvidia Pascal GPU with 256 CUDA cores
		Dual-core ARM Denver 2 CPU and quad-core ARM Cortex-A57 CPU
	Memory	8 GB LPDDR4
	Operating system	Nvidia JetPack SDK
	System clock	38.4 MHz
	Sleep clock	32.768 MHz

Fig. 15. Real-word CIDPS implementation.

requirements for intrusion detection and network monitoring. The Jetson AGX Orin Developer Kit is a high-performance platform tailored to handle the computational demands of UAV networks. The Jetson AGX Orin provides substantial computational power and capabilities suitable for real-time intrusion detection within a UAV network. The UAV network comprises a fleet of UAVs, each equipped with communication systems capable of transmitting network data to the Jetson AGX Orin Developer Kit. Specialties of UAV network devices and Jetson TX2s are presented in Table 13.

Additionally, a GCS acts as a central hub for monitoring and controlling the UAV network. The GCS communicates with the UAVs and interfaces with the Jetson AGX Orin for network data analysis, as shown in Fig. 15. The two primary functions carried out by the IDS deployed in this UAV network are online and offline attacks prediction. When operating in the online mode, the IDS continually records network packets from the GCS and UAVs in real-time. These packets undergo pre-processing to prepare them for input into the model. The IDS then generates predictions based on the inference results in real-time, aiding in the immediate detection of network intrusions.

In the offline prediction mode, the IDS conducts inferences on pre-processed datasets collected earlier. This mode is valuable for analyzing historical network data and identifying past intrusion patterns. Fig. 17 illustrates the execution time for online and offline prediction modes in this UAV networks IDS implementation. By combining online and offline prediction modes, the IDS benefits from both real-time monitoring and the knowledge gained from historical data. This hybrid approach allows for faster and more accurate intrusion detection because the system does not need to wait for a significant amount of historical data to accumulate before making predictions. The Jetson AGX Orin Developer Kit's exceptional computational capabilities ensure efficient and low-latency intrusion detection in the dynamic environment of UAV networks.

The real-time implementation exhibits a slightly slower execution time compared to the offline counterpart, primarily attributed to the pre-processing tier essential for live operation. Specifically, when employing a Jetson TX2, the trained model necessitates an average of 118 ms to perform a single inference. However, it is notable that contemporary automotive controllers typically possess more limited processing capabilities, making the execution of DL models challenging. To enable the seamless operation of this system in such contexts, the integration of an external computational device, such as a Jetson TX2 or equivalent hardware, becomes imperative.

6.4.1. CIDPS performance

The proposed CIDPS is designed to continuously classify sessions with each incoming packet until it determines the session's class type.

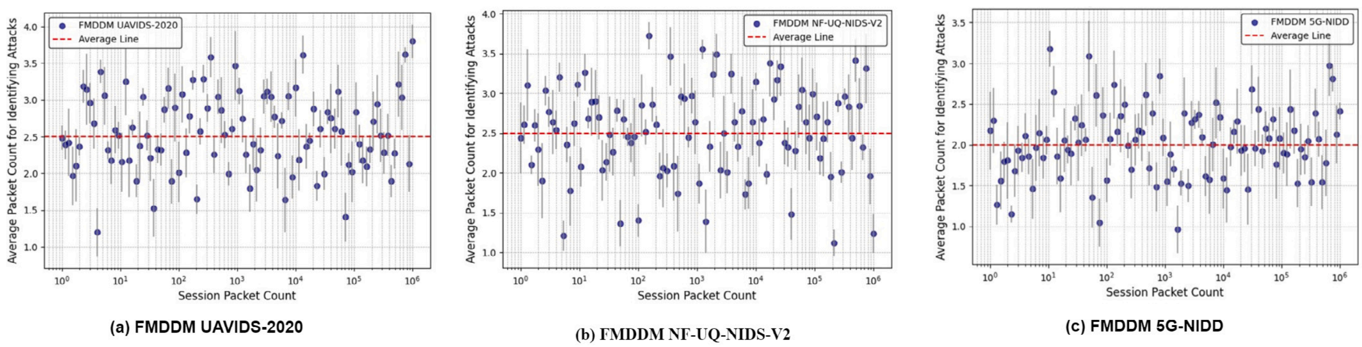

Fig. 16. Length of each session and the number of packets needed to detect an attack, with vertical lines showing the packet count range. (a) FMDDM UAVIDS-2020, (b) FMDDM NF-UQ-NIDS-V2, (c) FMDDM 5G-NIDD.

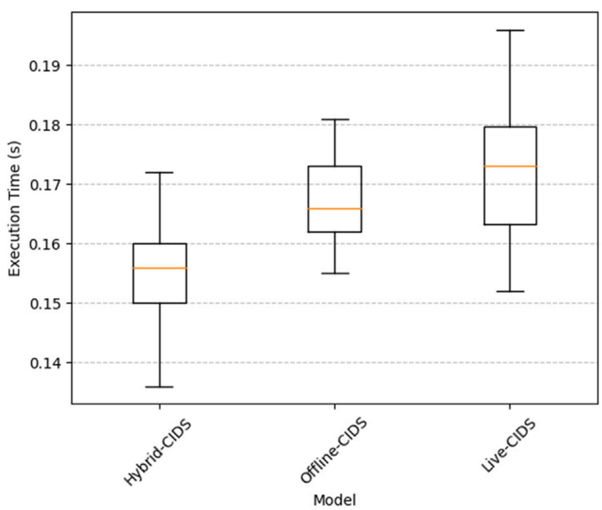

Fig. 17. Model execution time.

This approach differs from traditional session-based IDPS, which typically classifies a session just once. As a result, the proposed CIDPS undertakes numerous classifications, potentially increasing the system's overhead significantly when compared to conventional systems. This heightened overhead could pose challenges for real-time processing. Therefore, minimizing the number of packets required for each session's classification is essential to enhance the system's efficiency. In an in-depth examination of system performance, Fig. 16 presents a comparative analysis between the total packet volume within a session and the requisite packet count for initial detection. The graphical representation explains that, in most scenarios, the detection mechanism necessitates fewer than four packets to accurately identify potential threats. This efficiency is particularly noteworthy in sessions characterized by an extensive packet count (surpassing 1000), where the requisite number of packets for successful detection consistently maintains a minimal threshold, illustrating a negligible escalation even as the session's packet volume significantly expands.

However, the main findings from the real-world experiments demonstrate that the proposed ClDPS can effectively classify sessions as normal or under attack with minimal packet information. Specifically, when utilizing the FMDDM NF-UQ-NIDS-V2 dataset, the system is able to detect attacks using just the first two packets in sessions that contain more than 120,000 packets. This indicates a high efficiency of the system in maintaining low system load while ensuring accurate attack detection. Such performance emphasizes the potential of the proposed method for real-time security applications, as it successfully handles large volumes of data with minimal resource utilization. The effectiveness of the system, particularly with the dataset used, highlights its capability to operate efficiently under realistic and high-volume conditions.

Additionally, the implementation of a CIDPS in UAV networks faces significant limitations, primarily stemming from two key challenges. First, the variability in manufacturing standards among different UAV models presents substantial integration challenges. This variability, which includes differences in hardware capabilities, communication protocols, and software interfaces, necessitates extensive customization to ensure compatibility across diverse platforms. Such customization can significantly escalate costs and add complexity. Second, UAVs are typically constrained by limited computational power, battery life, and payload capacity. The additional demands of a CIDPS, such as increased computational requirements and enhanced communication needs, can adversely impact the operational efficiency and duration of UAV missions. Addressing these challenges is crucial for the successful deployment and functionality of CIDPS in UAV networks.

7. Conclusion

Cyberattacks on UAV-enabled 6G networks might have catastrophic effects, such as long-term environmental harm, deaths, and enormous financial losses. These infrastructures need specialized security solutions, ensuring that efficiency gains do not compromise security measures. To tackle this issue, the focus is on creating reliable IDS. However, the shortcomings in detection features often hamper the accuracy of detecting various emerging cyber-attacks. Therefore, our study identified specific features that can enhance the effectiveness of ML-based IDS. Special recommendations have been made to guard against DoS/DDoS, scanning, probing, and reconnaissance attacks, specifically for connection-based traffic features and time-based content in addition to packet headers. It includes a thorough analysis of many potential features in the dataset for the detection of attacks, such as zero-day attacks and known attacks. Moreover, our work proposes a real-time fusion multi-tier DNN model that provides reliable IDS for these crucial networks.

Next, the multi-tier DNN produces a final judgment with improved accuracy by combining the decisions made at each tier in a combination function. The highest accuracy achieved by our fusion multi-tier DNN model is 99.95%, which outperforms the existing models. This approach is significantly efficacious for 6G networks with UAV capability, where security is more important than latency or other productivity factors. It provides improved attack classification accuracy. Besides, this study also suggests a cooperative IPS that uses two multi-tier DNNs. One is for classifying the discovered anomalies and applying supplementary defensive measures. The other DNN is for detecting anomalies and acting quickly to take emergency defensive action. Additionally, the suggested deployment architecture uses network virtualization to provide CIDPS in UAV-enabled 6G networks.

CRediT authorship contribution statement

Hassan Jalil Hadi: Writing – original draft, Designing the study, Developing the research methodology, Conducting experiments. **Yue Cao:** Supervision, Review & editing, Revision and editing of the manuscript, Providing critical feedback and improvements. **Sifan Li:** Conceptualization, Methodology, Investigation. **Lexi Xu:** Data analysis, Visualization. **Yulin Hu:** Resources, Provided important resources or expertise for the research. **Mingxin Li:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yue Cao reports financial support was provided by Wuhan University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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